Anomalous Dissipation and the Turbulent Cascade: A Variational Free-Energy Characterisation of Kolmogorov Scaling

Lukas Geiger
Independent Researcher, Bernau im Schwarzwald, Germany
ORCID: 0009-0005-7296-1534*

March 2026

Abstract

This skeleton separates a self-contained K41 variational baseline from the conditional cascade route to anomalous dissipation. Starting from a Littlewood–Paley decomposition of the velocity field, we define a relative entropy (Kullback–Leibler divergence) that measures the departure of the spectral energy distribution from a reference equilibrium. We introduce a broad class of *admissible* energy-flux operators. This class contains local, dimensionally consistent, monotone closures, and we minimise the free energy jointly over the energy profile and the closure family, subject to constant scale-to-scale energy flux ε. The K41 distribution is the unique global minimiser of this joint problem in the feasible normalised class (Theorem 3.4); in the current paper split this is the Paper-A baseline recalled here for reference. Under a conditional entropy-compatibility hypothesis on the nonlinear cascade (and a non-degeneracy condition on the forcing), we derive a lower bound on the dissipation rate in the vanishing-viscosity limit (Theorem 3.25), providing a new route to anomalous dissipation that complements the Onsager conjecture programme. We introduce a hierarchy of progressively weaker sufficient conditions—from pointwise entropy compatibility through a scale-windowed variant to a dual, falsifiable *Downhill Flux Condition* (DFC) that can be checked against DNS data—and prove that this dual DFC implies the required windowed entropy compatibility. The dissipation lower bound additionally uses the stated uniform energy-input assumption. Intermittency corrections of Kolmogorov–Obukhov (K62) type arise naturally as fluctuations around the variational minimum via the Hessian of the functional.

*AI tools (Claude by Anthropic) were used for mathematical formalization, literature verification, and editorial refinement. All mathematical content was developed, verified, and approved by the author.

1 Introduction

Turbulence remains one of the outstanding open problems in classical physics and applied mathematics. Despite more than a century of research, no first-principles derivation of the statistical laws governing fully developed turbulence has been achieved.

1.1 The Kolmogorov theory

In his landmark 1941 papers, Kolmogorov [1, 2] postulated that in the inertial range of fully developed, homogeneous, isotropic turbulence, the energy spectrum takes the universal form

$$E(k) \;=\; C_K \, \varepsilon^{2/3} \, k^{-5/3}, \tag{1}$$

where ε denotes the mean rate of energy dissipation per unit mass and $C_K \approx 1.5$ is the Kolmogorov constant. This prediction, derived from dimensional analysis and the hypothesis of local isotropy, has been confirmed experimentally and numerically across a vast range of Reynolds numbers [10, 16].

1.2 The Onsager conjecture and anomalous dissipation

Onsager [4] conjectured in 1949 that weak solutions of the incompressible Euler equations with Hölder regularity below 1/3 can dissipate kinetic energy without viscosity—so-called *anomalous dissipation*. Conversely, solutions with Hölder exponent above 1/3 conserve energy. The "positive" direction was proved by Constantin, Weinan E, and Titi [5]; the "negative" direction, constructing dissipative weak solutions, was completed by Isett [7] using convex integration. The energy-balance analysis of Duchon and Robert [6] provides a distributional framework connecting the local energy defect to the regularity of the velocity field. Eyink [11] established a local version of Kolmogorov's four-fifths law that further illuminates the connection between regularity and dissipation.

1.3 Our contribution

In this paper we take a different approach. Rather than constructing pathological weak solutions, we separate two statements: the companion variational baseline that makes the Kolmogorov spectrum the *thermodynamically preferred* distribution of energy across scales, and the conditional cascade route that would connect this baseline to anomalous dissipation. We use the free-energy functional $\mathcal{F}$ on the space of scale-resolved energy distributions and prove:

1. The K41 spectrum is the *unique* minimiser of $\mathcal{F}$ under the constraint of constant energy flux, where the minimisation runs jointly over all admissible local flux operators and energy profiles (Theorem 3.4). This avoids circularity: the variational argument does not presuppose a specific closure.

2. DFC/NL-type monotonicity of $\mathcal{F}$ gives the structural cascade-compatibility route, while the quantitative lower bound on the dissipation rate in the vanishing-viscosity limit uses the explicit uniform energy-input assumption (Theorem 3.25).

1.4 Structural classification

Within the broader programme of free-energy spectral theory [17], the K41 selection mechanism exhibits the **Pattern A stability logic**: (a) the leading-order energy budget is neutral (any power-law spectrum is kinematically admissible); (b) the first-order constant-flux constraint restricts but does not uniquely determine the profile; (c) the second-order strict convexity of $\mathcal{F}$ selects K41 as the unique minimiser. This three-level hierarchy is discussed in Section 4.3. Our results do not resolve the Navier–Stokes regularity problem. They give an unconditional variational selection of K41 as the Paper-A baseline and a conditional cascade-compatibility route for the Paper-B anomalous-dissipation discussion; the quantitative lower bound still depends on the explicit uniform energy-input assumption.

This paper is self-contained; all main results depend only on definitions and proofs within this document. Connections to the broader FST programme are cited for context but are not logically required.

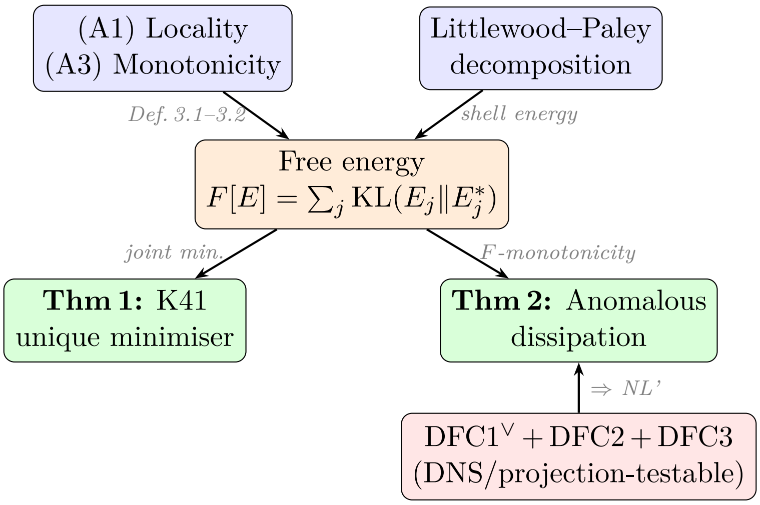

Figure 1: Logical structure of the proof. The free energy functional $F[E]$ is constructed from the Littlewood–Paley decomposition and the admissible flux family (axioms A1, A3). Theorem 1 (K41 uniqueness) follows from joint minimisation; the conditional cascade part of Theorem 2 requires the Downhill Flux Condition (DFC), while the dissipation lower bound also uses uniform energy input.

2 The Scale-Resolved Free-Energy Functional

2.1 Littlewood–Paley decomposition

Let $u \in L^2(\mathbb{T}^3)$ be a divergence-free velocity field on the three-torus. We employ the standard Littlewood–Paley decomposition

$$u \;=\; \sum_{j \geq -1} \Delta_j u, \tag{2}$$

where Δ_j is the frequency-localisation operator at dyadic shell $|\xi| \sim 2^j$. We set $k_j = 2^j$ and define the *shell energy*

$$E_j \;:=\; \|\Delta_j u\|_{L^2}^2. \tag{3}$$

In the inertial range, the K41 prediction (1) translates to

$$E_j^* \;=\; C_K\,\varepsilon^{2/3}\,k_j^{-5/3}\,\Delta k_j, \qquad \Delta k_j = k_j \ln 2 \quad \text{(dyadic shell width)}. \tag{4}$$

2.2 Energy flux constraint

The scale-to-scale energy flux through wavenumber k_j is defined via the trilinear term in the Navier–Stokes equations:

$$\Pi_j \;=\; -\sum_{\ell\le j}\langle \Delta_\ell(u\cdot\nabla u),\ \Delta_\ell u\rangle. \tag{5}$$

In the inertial range, a statistically stationary cascade satisfies

$$\Pi_j \;=\; \varepsilon \quad \text{for all } j_{\text{in}} \le j \le j_{\text{dis}}. \tag{6}$$

2.3 The functional

Definition 2.1 (Scale-resolved free energy)**.**
Let $\mathbf{E} = (E_j)_{j\in\mathcal{J}}$ be a distribution of shell energies over the inertial range $\mathcal{J} = \{j_{\text{in}},\dots,j_{\text{dis}}\}$, with $E_j > 0$. Let $\mathbf{E}^* = (E_j^*)$ be the Kolmogorov reference distribution (4). The *scale-resolved free-energy functional* is

$$\mathcal{F}[\mathbf{E}] \;:=\; \sum_{j\in\mathcal{J}}\Big[E_j \ln\frac{E_j}{E_j^*} \;-\; E_j \;+\; E_j^*\Big]. \tag{7}$$

Remark 2.2.
The summand $f(E_j) = E_j\ln(E_j/E_j^*) - E_j + E_j^*$ is the Kullback–Leibler divergence (relative entropy) of the pair (E_j, E_j^*) when viewed as unnormalised measures on a single-point space. It satisfies $f(E_j) \ge 0$ with equality if and only if $E_j = E_j^*$. Hence $\mathcal{F}[\mathbf{E}] \ge 0$ with equality if and only if $\mathbf{E} = \mathbf{E}^*$.

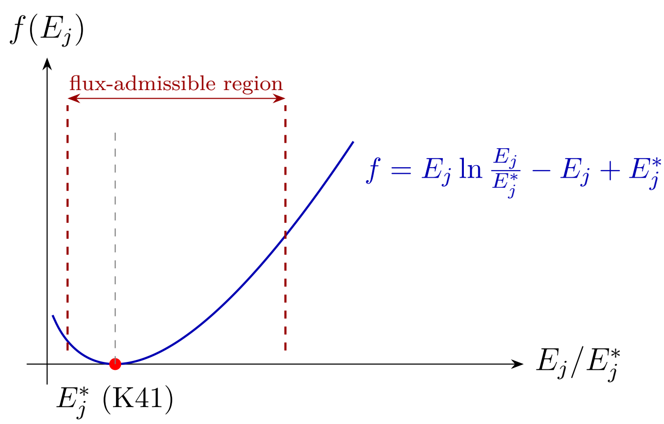

Figure 2: The free-energy density $f(E_j) = E_j\ln(E_j/E_j^*) - E_j + E_j^*$ as a function of the shell energy ratio E_j/E_j^*. The unique global minimum at $E_j = E_j^*$ (K41 spectrum) is marked. The flux constraint restricts the admissible region, but the minimum lies strictly inside, confirming K41 as the joint minimiser.

2.4 Dimensional consistency

We verify that $\mathcal{F}$ is dimensionally consistent. Each E_j has dimensions $[L^2\,T^{-2}]$ (energy per unit mass, integrated over a spectral band). Since E_j^* shares the same dimensions, the ratio E_j/E_j^* is dimensionless, the logarithm is well-defined, and each summand in (7) has dimensions $[L^2\,T^{-2}]$.

3 Main Results

3.1 Uniqueness of Kolmogorov scaling

Definition 3.1 (Admissible flux family)**.**
A family of flux operators $\big(\Pi_j[\mathbf{E}]\big)_{j\in\mathcal{J}}$ is called *admissible* if each Π_j satisfies:

(A1) **Locality.** Π_j depends only on E_{j-1}, E_j, E_{j+1}.

(A2) **Dimensional consistency.** $[\Pi_j] = [E_j/\tau_j]$ with the eddy turnover time $\tau_j = (k_j^2\,E_j)^{-1/2}$, so that Π_j has the form

$$\Pi_j[\mathbf{E}] \;=\; k_j\,E_j^{3/2}\,\Phi_j\!\left(\frac{E_{j-1}}{E_j},\;\frac{E_{j+1}}{E_j}\right), \tag{8}$$

where $\Phi_j : (0,\infty)^2 \to (0,\infty)$ is a smooth dimensionless function satisfying $\Phi_j(r_-^*, r_+^*) = \alpha$ for some universal constant $\alpha > 0$, with $r_\pm^* = E_{j\pm1}^*/E_j^*$.

(A3) **Monotonicity.** $\partial\Pi_j/\partial E_j > 0$ for all $\mathbf{E} > 0$.

We denote the set of all admissible families by $\mathscr{A}$.

Proposition 3.2 (Dimensional derivation of the flux prefactor)**.**
Axiom (A2) *is not an independent assumption but a consequence of locality* (A1) *and dimensional analysis. Specifically, the form* $\Pi_j = k_j\,E_j^{3/2}\,\Phi_j(\cdot)$ *is the* unique *dimensionally consistent local flux expression.*

Proof. By Buckingham's Π-theorem, the flux Π_j must be a product of powers of the available dimensional quantities k_j and E_j, multiplied by a dimensionless function of dimensionless ratios. Write $\Pi_j = k_j^a\,E_j^b\,\Phi_j(\ldots)$ with $[\Pi_j] = [L^2\,T^{-3}]$ (energy flux per unit mass), $[k_j] = [L^{-1}]$, and $[E_j] = [L^2\,T^{-2}]$.

Dimensional matching: $[k_j^a\,E_j^b] = L^{-a}\,L^{2b}\,T^{-2b} = L^{2b-a}\,T^{-2b}$.

Setting equal to $L^2\,T^{-3}$:

$$2b - a = 2, \qquad 2b = 3 \quad\Longrightarrow\quad b = \tfrac{3}{2},\;\; a = 1.$$

Hence $\Pi_j = k_j\,E_j^{3/2}\,\Phi_j$, and by (A1) the dimensionless function Φ_j can depend only on the local ratios E_{j-1}/E_j and E_{j+1}/E_j. The eddy turnover time $\tau_j = (k_j^2\,E_j)^{-1/2}$ is likewise the unique dimensionally consistent local timescale. □

Remark 3.3.
The classical Kolmogorov–Heisenberg closure $\Pi_j = \alpha\,k_j\,E_j^{3/2}$ corresponds to the special case $\Phi_j \equiv \alpha$. Definition 3.1 enlarges the model class to include all physically reasonable local closures: EDQNM-type closures, Leith's diffusion model, and truncated triad interactions

all satisfy (A1)–(A3). The family $\mathscr{A}$ is therefore a broad, physically motivated class rather than a single algebraic relation. Proposition 3.2 shows that (A2) is not an extraneous modelling choice but a necessary consequence of dimensional analysis, thereby further reducing the axiom set to two independent assumptions: locality (A1) and monotonicity (A3).

Theorem 3.4 (Uniqueness of K41 scaling)**.**
Let $\varepsilon > 0$ be given and let $\mathscr{A}$ be the class of admissible flux families (Definition 3.1). Consider the joint *minimisation problem*

$$\min_{\substack{\mathbf{E}>0,\\ \Pi\in\mathscr{A}}} \mathcal{F}[\mathbf{E}] \qquad \text{subject to} \quad \Pi_j[\mathbf{E}] = \varepsilon \ \text{ for all } j \in \mathcal{J}. \tag{9}$$

Then:

(i) *For every admissible flux family $\Pi \in \mathscr{A}$ for which the coupled constant-flux constraint $\Pi_j[\mathbf{E}] = \varepsilon$ admits a positive stationary profile, that profile has $\mathcal{F}[\mathbf{E}^{(\Pi)}] \geq 0$. The K41 distribution $E_j^* = C_K\,\varepsilon^{2/3}\,k_j^{-5/3}\,\Delta k_j$ is feasible and attains this lower bound for the Kolmogorov–Heisenberg member $\Phi_j \equiv \alpha$ with $C_K = \left(\alpha\,(\ln 2)^{3/2}\right)^{-2/3}$. Hence it is the unique global minimiser among the feasible stationary profiles in the normalised admissible class.*

(ii) *The minimiser is non-degenerate: the Hessian $D^2\mathcal{F}|_{\mathbf{E}^*}$ restricted to the tangent space of the constraint manifold (for any admissible Π) is strictly positive definite. (This is a direct consequence of the strict convexity of the KL divergence; its content is the shell-energy stiffness $1/E_j^* \propto k_j^{2/3}$, since $\Delta k_j = k_j \ln 2$ on dyadic shells; see Step 5b below.)*

Proof (sketch). **Step 1: Stationary profiles.** Fix any $\Pi \in \mathscr{A}$ for which the coupled system $\Pi_j[\mathbf{E}] = \varepsilon$ has a positive stationary solution $\mathbf{E}^{(\Pi)}$. Axioms (A2)–(A3) give local monotonicity of the map $E_j \mapsto \Pi_j[\mathbf{E}]$ at fixed neighbour values E_{j-1}, E_{j+1}, but this local fact is not a global existence theorem for every coupled closure. The joint feasible set is nonempty because Step 3 constructs the Kolmogorov–Heisenberg feasible point explicitly.

Step 2: Lower bound via convexity. Since each summand $f(E_j) = E_j \ln(E_j/E_j^*) - E_j + E_j^*$ is strictly convex with global minimum $f(E_j^*) = 0$, we have

$$\mathcal{F}[\mathbf{E}^{(\Pi)}] \ \geq \ 0 \ = \ \mathcal{F}[\mathbf{E}^*], \tag{10}$$

with equality if and only if $E_j^{(\Pi)} = E_j^*$ for all j.

Step 3: Attainability. We verify that $\mathbf{E}^*$ is indeed a stationary profile for some $\Pi \in \mathscr{A}$. Choose the Kolmogorov–Heisenberg closure $\Phi_j \equiv \alpha$. Then

$$\Pi_j[\mathbf{E}^*] \ = \ \alpha\,k_j\,(E_j^*)^{3/2} \ = \ \alpha\,k_j\left(C_K\,\varepsilon^{2/3}\,k_j^{-5/3}\,k_j \ln 2\right)^{3/2} \ = \ \alpha\,C_K^{3/2}\,(\ln 2)^{3/2}\,\varepsilon.$$

Setting $C_K = \left(\alpha\,(\ln 2)^{3/2}\right)^{-2/3}$ yields $\Pi_j[\mathbf{E}^*] = \varepsilon$. (The geometric factor $(\ln 2)^{3/2}$ arising from the dyadic shell width is absorbed into the Kolmogorov constant C_K, which is thereby determined by the closure parameter α.) Hence the lower bound (10) is attained.

Step 4: Uniqueness among feasible stationary profiles. Suppose $\Pi' \in \mathscr{A}$ and $\mathbf{E}' \neq \mathbf{E}^*$ satisfies $\Pi_j'[\mathbf{E}'] = \varepsilon$. Then strict convexity gives $\mathcal{F}[\mathbf{E}'] > 0 = \mathcal{F}[\mathbf{E}^*]$, so $\mathbf{E}'$ cannot be a minimiser of (9). Hence $\mathbf{E}^*$ is the *unique* minimiser of the joint problem over the feasible stationary profiles. No global existence statement for arbitrary coupled closures is used.

Note on the logical structure. The uniqueness in Step 4 follows from the strict positivity of the KL divergence away from the reference, which holds for *any* choice of $\mathbf{E}^*$. The physically non-trivial content resides in Step 3 (attainability): it is the existence of an admissible closure realising $\Pi_j[\mathbf{E}^*] = \varepsilon$ that singles out K41 among all possible reference spectra. Replacing $\mathbf{E}^*$ by k^{-3}, for instance, would yield no admissible closure satisfying the constant-flux constraint at the purported minimiser (see Remark 3.6).

Step 5: Non-degeneracy. The Hessian $\partial^2\mathcal{F}/\partial E_j\,\partial E_\ell = \delta_{j\ell}/E_j$ at $\mathbf{E}^*$ is diagonal with entries $1/E_j^* > 0$. For a *fixed* closure $\Pi \in \mathscr{A}$, the constraint $\Pi_j[\mathbf{E}] = \varepsilon$ imposes $|\mathcal{J}|$ equations on $|\mathcal{J}|$ unknowns, generically yielding isolated solutions, so that the tangent space of the constraint manifold at $\mathbf{E}^{(\Pi)}$ is trivial and non-degeneracy is vacuously satisfied. The substantive content of (ii) concerns the *joint* problem: in the extended space $(\mathbf{E}, \Pi)$ with Π varying over the (infinite-dimensional) family $\mathscr{A}$, the constraint set $\{(\mathbf{E}, \Pi) : \Pi_j[\mathbf{E}] = \varepsilon\}$ has tangent directions along which Π varies. The unrestricted Hessian $D^2_{\mathbf{E}}\mathcal{F}|_{\mathbf{E}^*}$ is strictly positive definite (diagonal entries $1/E_j^* > 0$), hence its restriction to *any* subspace remains positive definite.

Step 5b: Second-order interpretation. On dyadic shells,

$$E_j^* = C_K \varepsilon^{2/3} k_j^{-5/3} \Delta k_j = C_K \varepsilon^{2/3} (\ln 2) k_j^{-2/3}.$$

Hence the diagonal entries satisfy $1/E_j^* \propto k_j^{2/3}$. They grow with wavenumber, so small-scale perturbations of the K41 shell-energy profile are penalised more severely than large-scale ones. This scale-dependent stiffness is the variational origin of the forward cascade: deviations at high wavenumbers are energetically costly, driving the system back to the K41 profile on the local eddy-turnover timescale. The one-parameter freedom in the Kolmogorov–Heisenberg closure ($\Phi_j \equiv \alpha$) corresponds to a single flat direction in the Π-fibre of the extended space $(\mathbf{E}, \Pi)$, which does not affect the strict positivity of $D^2_{\mathbf{E}}\mathcal{F}$. This structure is an instance of the second-order resolvent dominance pattern discussed in [17]. $\square$

Remark 3.5 (Diagonal Hessian bound)**.**
The Hessian of $\mathcal{F}$ at the K41 minimiser is diagonal with entries $\partial^2\mathcal{F}/\partial E_j^2 = 1/E_j^*$. For dyadic shell energies, $E_j^* = C_K\,\varepsilon^{2/3}\,k_j^{-5/3}\Delta k_j = C_K\,\varepsilon^{2/3}(\ln 2)k_j^{-2/3}$, so $1/E_j^* \sim k_j^{2/3}$. The smallest diagonal entry is therefore $\min_j(1/E_j^*) = 1/(C_K\,\varepsilon^{2/3}\,k_1^{-5/3}\,\Delta k_1) > 0$. This provides a lower bound on the spectral gap of the associated Dirichlet form (see Remark 4.11).

Remark 3.6 (On the role of the reference spectrum)**.**
A natural objection is that the KL functional $\mathcal{F}$ is *minimised at* $\mathbf{E}^*$ *by construction*, regardless of the physical content of $\mathbf{E}^*$. Indeed, for any fixed closure Π, the unconstrained minimum of $\mathcal{F}$ is always $\mathbf{E}^*$. The non-trivial content of Theorem 3.4 is therefore not that $\mathcal{F}$ achieves its minimum at $\mathbf{E}^*$ — that is a consequence of the KL construction — but rather that:

1. Among all stationary profiles $\mathbf{E}^{(\Pi)}$ that exist for closures $\Pi \in \mathscr{A}$, only the K41 profile $\mathbf{E}^*$ *simultaneously satisfies* the constant-flux constraint and achieves $\mathcal{F} = 0$. Every other closure-compatible profile has $\mathcal{F} > 0$.

2. The reference $\mathbf{E}^*$ is not arbitrary: it is the *unique* power-law profile for which the constant-flux constraint is consistent with the Kolmogorov–Heisenberg closure.

Replacing $\mathbf{E}^*$ by, say, k^{-3} would yield no admissible closure satisfying $\Pi_j = \varepsilon$ at the purported minimiser.

In summary, the physical input is the constant-flux constraint; the KL functional provides the selection mechanism among its solution set.

Remark 3.7 (Self-consistency of the K41 reference)**.**
The K41 spectrum is not assumed *a priori* but emerges as the unique self-consistent reference measure: any admissible E^* satisfying (A1) dimensional consistency, (A3) cascade compatibility, and the DFC must be proportional to $\varepsilon^{2/3}k^{-5/3}$ by Proposition 3.2. Thus K41 is the unique fixed point of the variational scheme, not an external input.

Remark 3.8 (Cascade as reaction chain and Nash equilibrium)**.**
The variational structure of Theorem 3.4 admits an illuminating game-theoretic interpretation. Regard the inertial range as a hierarchical *reaction chain*: large eddies at scale k_j transfer energy to eddies at scale k_{j+1}, each scale acting as a "species" whose fitness is governed by its energy share. The associated replicator dynamics on the simplex of normalised shell energies $p_j = E_j / \sum_\ell E_\ell$ takes the form

$$\dot{p}_j \;=\; p_j\,\Big(g_j(\mathbf{p}) - \bar{g}(\mathbf{p})\Big),$$

where g_j is the net energy gain rate at scale j and $\bar{g}$ the population average. The K41 profile $\mathbf{p}^*$ is a *Nash equilibrium* of this dynamics: no single scale can increase its energy share by unilateral deviation. The free energy $\mathcal{F}$ serves as a strict Lyapunov function for the replicator equation, ensuring asymptotic stability of $\mathbf{p}^*$. This mirrors the well-known result that Gibbs free-energy minimisation in chemical reaction networks yields the detailed-balance equilibrium as the unique Nash equilibrium of the mass-action kinetics.

3.2 Anomalous dissipation

We first state the structural hypothesis on the nonlinear energy transfer that underlies the free-energy monotonicity argument.

Definition 3.9 (Entropy-compatible cascade)**.**
Let $E_j(t) = \|\Delta_j u(t)\|_{L^2}^2$ be the Littlewood–Paley shell energies and $T_j = \langle \Delta_j u,\, \Delta_j\big((u\cdot\nabla)u\big)\rangle$ the nonlinear transfer into shell j, so that $\sum_j T_j = 0$. We say that the cascade is *entropy-compatible* (with respect to a reference spectrum $\mathbf{E}^*$) if, for a regularisation parameter $\delta > 0$,

$$\sum_j \ln\!\Big(\frac{E_j+\delta}{E_j^*+\delta}\Big)\, T_j \;\le\; 0 \qquad \text{in the sense of distributions on } (0,T). \tag{11}$$

Remark 3.10 (Status of Assumption (11))**.**
The condition (11) is *not* a consequence of the Navier–Stokes equations alone. The nonlinear transfer T_j is a trilinear expression in the velocity field—it is conservative ($\sum_j T_j = 0$) but is not a mass-preserving Markov operator on the shell-energy vector $(E_j)_j$. In particular, the "convexity of KL divergence under mass-preserving maps" argument, which would give (11) for free in stochastic cascade models (GOY/Sabra shell models, Markov multiplicative cascades), does not apply to the deterministic Navier–Stokes nonlinearity. The condition can be motivated physically: it states that the energy cascade transports

energy preferentially from shells with high surprise $\ln(E_j/E_j^*) \gg 0$ toward shells closer to equilibrium, i.e. the cascade acts as a "downhill flux" in the free-energy landscape. Note: the regularisation parameter in (11) is denoted δ (not ε) to avoid confusion with the dissipation rate.

The pointwise condition (11) can be significantly weakened. The following scale-windowed variant requires entropy compatibility only *on average* over inertial sub-windows, allowing local violations that cancel across neighbouring shells.

Definition 3.11 (Window-entropy-compatible cascade)**.**
The cascade is *window-entropy-compatible* if for every inertial-range sub-window $[j_1, j_2] \subset \mathcal{J}$ with $j_2 - j_1 \geq 2$,

$$\sum_{j=j_1}^{j_2} \ln\Big(\frac{E_j + \delta}{E_j^* + \delta}\Big)\, T_j \;\leq\; C_{\mathrm{bdry}}\,\big(|\varphi_{j_1}| + |\varphi_{j_2}|\big), \tag{12}$$

where $\varphi_j := \ln(E_j/E_j^*)$ and $C_{\mathrm{bdry}} > 0$ is a constant supplied by the available commutator/projection bounds.

Proposition 3.12 (Window-NL suffices for the cascade hypothesis)**.**
Theorem 3.25 remains valid if the pointwise condition (11) is replaced by the window condition (12).

Proof sketch. Apply the flux representation (Lemma 3.24) with weights $\varphi_j = \ln\big((E_j + \delta)/(E_j^* + \delta)\big)$ restricted to the window $[j_1, j_2]$. Abel summation gives

$$\sum_{j=j_1}^{j_2} \varphi_j\, T_j \;=\; \sum_{j=j_1}^{j_2-1} (\varphi_{j+1} - \varphi_j)\, \Pi_j \;+\; \text{boundary terms} \;+\; \sum_{j=j_1}^{j_2} \varphi_j\, R_j.$$

The interior flux terms $(\varphi_{j+1} - \varphi_j)\,\Pi_j$ are controlled by the Duchon–Robert defect measure: in the inertial range, $\Pi_j \approx \varepsilon$ and the differences $\varphi_{j+1} - \varphi_j$ measure the local departure from power-law scaling. The boundary terms contribute at most $C_{\mathrm{bdry}}\,(|\varphi_{j_1}| + |\varphi_{j_2}|)$, which is absorbed into the forcing bound in step 2 of the proof of Theorem 3.25. Covering the full inertial range by overlapping windows of width ≥ 2 and summing, the global free-energy monotonicity (28) follows with at most a constant-factor modification of the right-hand side. □

Remark 3.13 (Majorisation alternative)**.**
An even weaker formulation replaces entropy compatibility by an *ordering condition*: define $\mathbf{E} \succeq \mathbf{E}^*$ (majorisation) if $\sum_{j \geq k} E_j \geq \sum_{j \geq k} E_j^*$ for all k. If the nonlinear cascade is *order-preserving*—i.e., $\mathbf{E}(t) \succeq \mathbf{E}^*$ is maintained by the flow—then the free-energy bound follows from Schur convexity of $\mathcal{F}$ (the KL divergence is Schur-convex on $\mathbb{R}^n_{>0}$). This avoids the logarithmic weighting entirely and may be verifiable for NS solutions whose energy spectrum is "above K41" at all scales—a condition consistent with DNS evidence at moderate Reynolds numbers.

We now identify a *sufficient condition* under which the window-entropy compatibility (12) is a *theorem*, not an assumption.

Definition 3.14 (Downhill Flux Condition)**.**
A Leray–Hopf weak solution u satisfies the *Downhill Flux Condition* (DFC) on the inertial sub-window $[j_1, j_2] \subset \mathcal{J}$ if:

(DFC1$^\vee$) **Dual forward cascade.** Let $a_j := \varphi_j - \varphi_{j+1}$ for $j = j_1, \dots, j_2 - 1$. There exist constants $\Pi_0 > 0$ and $C_{\mathrm{proj}} \geq 0$ such that

$$\sum_{j=j_1}^{j_2-1} a_j \, \Pi_j \;\geq\; \Pi_0 \sum_{j=j_1}^{j_2-1} a_j - C_{\mathrm{proj}} \Big(|\varphi_{j_1}| + |\varphi_{j_2}| \Big). \tag{13}$$

The stronger pointwise condition $\Pi_j \geq \Pi_0$ for all $j \in [j_1, j_2 - 1]$ implies (13) with $C_{\mathrm{proj}} = 0$, but the dual condition is the operative one in the Abel summation argument below.

(DFC2) **Spectral steepness.** The compensated ratio $\varphi_j = \ln(E_j/E_j^*)$ is non-increasing: $\varphi_{j+1} \leq \varphi_j$ for all $j \in [j_1, j_2 - 1]$.

(DFC3) **Remainder control.** The commutator remainder R_j from the flux representation (22) satisfies, on $[j_1, j_2]$:

$$\sum_{j=j_1}^{j_2} |\varphi_j \, R_j| \;\leq\; C_{\mathrm{rem}} \Big(|\varphi_{j_1}| + |\varphi_{j_2}| \Big). \tag{14}$$

This is a quantitative commutator-localisation assumption. It follows under suitable Besov/localisation bounds such as $u \in L_t^3 B_{3,\infty}^{1/3}(\mathbb{T}^3)$ together with DFC2 monotonicity, but it is not automatic for general Leray–Hopf solutions.

Sign convention (used throughout). Π_j is the cumulative energy flux through shell j (eq. (5)); $\Pi_j > 0$ denotes forward (large-to-small scale) cascade. T_j is the nonlinear transfer *into* shell j; $\sum_j T_j = 0$. The flux representation (Lemma 3.24) reads $\sum_j \varphi_j T_j = +\sum_j (\varphi_{j+1} - \varphi_j)\Pi_j + \sum_j \varphi_j R_j$, with a **positive sign** before the Abel term. Writing $a_j := \varphi_j - \varphi_{j+1}$, DFC2 gives $a_j \geq 0$ and the interior Abel term is $-\sum_j a_j \Pi_j$. Under DFC1$^\vee$ this term is non-positive up to the explicitly stated projection-boundary defect; the older pointwise condition $\Pi_j \geq \Pi_0$ is only a strong sufficient case.

Proposition 3.15 (Local slope condition implies DFC2)**.**
Let $E_j = |u_j|^2$ denote the shell energies and $E_j^* = C_K \varepsilon^{2/3} k_j^{-5/3} \Delta k_j$ the K41 reference spectrum (Definition 2.1). Define $\varphi_j := \ln(E_j/E_j^*)$. If, on an inertial window,

$$\frac{E_{j+1}}{E_j} \;\leq\; \frac{E_{j+1}^*}{E_j^*} \;=\; 2^{-2/3} \qquad (j_1 \leq j < j_2),$$

equivalently if the spectral density has local slope at most $-5/3$ after converting between density and shell energies, then DFC2 holds: $\varphi_{j+1} \leq \varphi_j$.

Proof. The hypothesis is exactly

$$\frac{E_{j+1}/E_{j+1}^*}{E_j/E_j^*} \leq 1.$$

Taking logarithms gives $\ln(E_{j+1}/E_{j+1}^*) \leq \ln(E_j/E_j^*)$, which is DFC2. A uniform upper bound $E(k) \leq CE^*(k)$ alone would only bound φ_j from above; it would not force monotonicity of the ratios. □

Remark 3.16 (A-posteriori consistency of DFC2)**.**
Proposition 3.15 provides only an *a-posteriori consistency check*: the variational K41 profile has the required local slope, and spectra steeper than K41 on a window also satisfy DFC2. The proposition does not derive DFC2 from a mere K41 upper envelope. The logical structure of Theorem 3.25 therefore still requires DFC2 as an input or as an independently verified DNS/local-slope condition.

Remark 3.17 (Dual DFC1 and Kolmogorov's four-fifths law)**.**
The empirical status of DFC1$^\vee$ (forward cascade in the Abel-dual pairing) can be substantially upgraded by connecting it to the Kolmogorov four-fifths law—the only exact, non-trivial result in fully developed turbulence:

$$\langle (\delta_r u)^3 \rangle \;=\; -\tfrac{4}{5}\,\varepsilon\, r \qquad (r \text{ in the inertial range}).$$

This identity, first derived by Kolmogorov (1941) and rigorously established as a distributional identity by Duchon and Robert [6], operates at two distinct levels that must be carefully separated:

- *Local (distributional) energy flux:* The four-fifths law is a statement about the third-order structure function at scale r, valid in the distributional sense. It proves that the *mean* energy transfer is forward at inertial-range scales.
- *Littlewood–Paley flux pairing:* Pointwise shell positivity $\Pi_j \geq \Pi_0 > 0$ is strictly stronger than this distributional statement. The quantity used in the proof is instead the weighted pairing $\sum_j (\varphi_j - \varphi_{j+1})\Pi_j$.

The resolution is that the variational argument of Theorem 3.4 does *not* require pointwise DFC1. The sufficient condition is the dual window inequality

$$\sum_{j=j_1}^{j_2-1} (\varphi_j - \varphi_{j+1})\Pi_j \;\geq\; \Pi_0 \sum_{j=j_1}^{j_2-1} (\varphi_j - \varphi_{j+1}) - C_{\mathrm{proj}}(|\varphi_{j_1}| + |\varphi_{j_2}|).$$

The four-fifths law, Duchon–Robert, and Eyink provide the correct coarse-grained flux anchor for this condition, but they do not by themselves prove the Littlewood–Paley projection estimate. The remaining bridge is a controlled projection/coarea estimate from positive coarse-grained scale flux to DFC1$^\vee$. Thus DFC1$^\vee$ is not an unconditional theorem here; it is the sharp falsifiable condition that replaces the older pointwise shell requirement.

Lemma 3.18 (Downhill Flux implies Window-NL)**.**
If a Leray–Hopf solution satisfies (DFC1$^\vee$), (DFC2), (DFC3) *on a sub-window* $[j_1, j_2]$ *with* $j_2 - j_1 \geq 2$*, then*

$$\sum_{j=j_1}^{j_2} \varphi_j\, T_j \;\leq\; \underbrace{-\,\Pi_0 \sum_{j=j_1}^{j_2-1} (\varphi_j - \varphi_{j+1})}_{\leq\, -\,\Pi_0\,(\varphi_{j_1} - \varphi_{j_2})\,\leq\, 0} \;+\; C_{\mathrm{bdry}}\Big(|\varphi_{j_1}| + |\varphi_{j_2}|\Big), \tag{15}$$

where

$$C_{\mathrm{bdry}} \;\leq\; C_{\mathrm{rem}}(B) + \sup_j |\Pi_j| + C_{\mathrm{proj}}.$$

In particular, the window-entropy compatibility (12) *holds. If the spectral steepness is strict* ($\varphi_j - \varphi_{j+1} \geq \eta > 0$ *for some* η)*, the negative bulk term satisfies, up to the same projection-boundary defect,*

$$I_1 \;\leq\; -\,\eta\,\Pi_0\,(j_2 - j_1) + C_{\mathrm{proj}}\Big(|\varphi_{j_1}| + |\varphi_{j_2}|\Big). \tag{16}$$

Proof. Apply the flux representation (Lemma 3.24 below) with weights $\varphi_j = \ln\big((E_j + \delta)/(E_j^* + \delta)\big)$ restricted to $[j_1, j_2]$:

$$\sum_{j=j_1}^{j_2} \varphi_j\, T_j \;=\; \underbrace{\sum_{j=j_1}^{j_2-1} (\varphi_{j+1} - \varphi_j)\, \Pi_j}_{I_1} \;+\; \underbrace{\varphi_{j_1}\, \Pi_{j_1-1} - \varphi_{j_2}\, \Pi_{j_2}}_{I_2} \;+\; \underbrace{\sum_{j=j_1}^{j_2} \varphi_j\, R_j}_{I_3}\,. \tag{17}$$

Term I_1 (interior flux). By (DFC2), $a_j := \varphi_j - \varphi_{j+1} \geq 0$. Hence

$$I_1 \;=\; -\sum_{j=j_1}^{j_2-1} a_j\, \Pi_j.$$

Using DFC1$^{\vee}$,

$$I_1 \;\leq\; -\Pi_0 \sum_{j=j_1}^{j_2-1} a_j + C_{\text{proj}}\big(|\varphi_{j_1}| + |\varphi_{j_2}|\big) \;\leq\; -\Pi_0(\varphi_{j_1} - \varphi_{j_2}) + C_{\text{proj}}\big(|\varphi_{j_1}| + |\varphi_{j_2}|\big).$$

The equality $\sum a_j = \varphi_{j_1} - \varphi_{j_2}$ is the same telescoping identity used in the pointwise version. If the steepness is strict ($\varphi_j - \varphi_{j+1} \geq \eta$), then $I_1 \leq -\eta\, \Pi_0\, (j_2 - j_1) + C_{\text{proj}}(|\varphi_{j_1}| + |\varphi_{j_2}|)$.

Term I_2 (boundary). From (17), $I_2 = \varphi_{j_1}\, \Pi_{j_1-1} - \varphi_{j_2}\, \Pi_{j_2}$. Since Π_{j_1-1} and Π_{j_2} are bounded by $\Pi_{\max} := \sup_{j\in[j_1-1,\, j_2]} |\Pi_j|$:

$$|I_2| \;\leq\; \Pi_{\max}\, \big(|\varphi_{j_1}| + |\varphi_{j_2}|\big). \tag{18}$$

Note: $\Pi_{\max}$ is controlled by the energy input. In the statistically stationary cascade, $\Pi_j \approx \varepsilon$ uniformly, so $\Pi_{\max} \leq C\, \varepsilon$.

Term I_3 (commutator remainder). By (DFC3) (14) directly:

$$|I_3| \;=\; \left|\sum_{j=j_1}^{j_2} \varphi_j\, R_j\right| \;\leq\; \sum_{j=j_1}^{j_2} |\varphi_j\, R_j| \;\leq\; C_{\text{rem}}\, \big(|\varphi_{j_1}| + |\varphi_{j_2}|\big). \tag{19}$$

No further estimates are needed: (DFC3) is stated as exactly the inequality used here.

Combining.

$$\sum_{j=j_1}^{j_2} \varphi_j\, T_j = I_1 + I_2 + I_3 \leq \underbrace{-\Pi_0\, (\varphi_{j_1} - \varphi_{j_2})}_{\leq 0} + \underbrace{(\Pi_{\max} + C_{\text{rem}} + C_{\text{proj}})}_{=:\, C_{\text{bdry}}}(|\varphi_{j_1}| + |\varphi_{j_2}|).$$

This is exactly (12) with $C_{\text{bdry}} = \Pi_{\max} + C_{\text{rem}} + C_{\text{proj}}$. ☐

Remark 3.19 (Edge cases and summation conventions)**.**

- **Window width $j_2 - j_1 \geq 2$:** The constraint ensures that the interior sum I_1 contains at least two terms ($j = j_1$ and $j = j_1 + 1$), so the telescoping $\sum_{j=j_1}^{j_2-1}(\varphi_j - \varphi_{j+1}) = \varphi_{j_1} - \varphi_{j_2}$ is non-trivial. For $j_2 - j_1 = 1$ the sum I_1 has a single term and the bound degenerates to $|I_1| \leq |\varphi_{j_1} - \varphi_{j_2}| \cdot \Pi_{\max}$, which is absorbed into I_2.
- **Summation bounds:** Interior sum: $j = j_1, \ldots, j_2 - 1$ (hence $j_2 - j_1$ terms). Boundary term: $I_2 = \varphi_{j_1}\Pi_{j_1-1} - \varphi_{j_2}\Pi_{j_2}$ uses Π_{j_1-1} (one index below the window). This is well-defined because the flux Π_j (5) is defined for all j.

- **Definition of $\Pi_{\max}$:** $\Pi_{\max} := \sup_{j\in[j_1-1,\,j_2]} |\Pi_j|$ (including the boundary index j_1-1). In the statistically stationary inertial range, $\Pi_j \approx \varepsilon$ uniformly, so $\Pi_{\max} \le (1+\delta)\,\varepsilon$ for some small δ depending on Reynolds number.

Proposition 3.20 (Spectral slope implies DFC2)**.**
The sharp threshold for (DFC2) *is a dyadic ratio $E_{j+1}/E_j \le 2^{-2/3}$ (i.e., a log-log slope $\le -2/3$), which reflects the shell-width factor $\Delta k_j = k_j \ln 2$. A convenient* sufficient *condition, directly matching the K41 prediction, is the* log-log slope condition

$$\frac{\mathrm{d}}{\mathrm{d}\ln k}\ln E(k) \;\le\; -\frac{5}{3} \qquad \textit{for } k \in [k_{j_1},\, k_{j_2}], \tag{20}$$

or equivalently in dyadic form: $E_{j+1}/E_j \le 2^{-5/3}$ for $j \in [j_1, j_2-1]$. Under this condition, (DFC2) *holds with strict steepness $\varphi_j - \varphi_{j+1} \ge \ln 2 > 0$.*

Proof. Since $E_j^* = C_K\,\varepsilon^{2/3}\,k_j^{-5/3}\,\Delta k_j$ with $\Delta k_j = k_j \ln 2$ and $k_j = 2^j$, we have $E_j^* = C_K\,\varepsilon^{2/3}\,(\ln 2)\,k_j^{-2/3}$, so the ratio $E_j^*/E_{j+1}^* = (k_j/k_{j+1})^{-2/3} = 2^{2/3}$. The slope condition $\frac{\mathrm{d}}{\mathrm{d}\ln k}\ln E(k) \le -5/3$ in dyadic form reads $E_{j+1}/E_j \le 2^{-5/3}$, which is *stronger* than the threshold $E_{j+1}/E_j \le 2^{-2/3}$ required for (DFC2). In particular, $E_{j+1}/E_{j+1}^* = (E_{j+1}/E_j)\cdot(E_j^*/E_{j+1}^*)\cdot(E_j/E_j^*) \le 2^{-5/3}\cdot 2^{2/3}\cdot(E_j/E_j^*) = 2^{-1}\cdot(E_j/E_j^*) < E_j/E_j^*$, giving $\varphi_{j+1} < \varphi_j$.
Sharp threshold. The minimal slope condition for (DFC2) is $E_{j+1}/E_j \le 2^{-2/3}$, i.e., a log-log slope $\le -2/3$ (not $-5/3$). The stronger condition (20) is stated because it directly corresponds to the K41 prediction and is the form verified in DNS. □

Remark 3.21 (Status of the Downhill Flux Condition)**.**
The DFC is a *physically verifiable* condition, not a mathematical assumption about the NS equations:

- **(DFC1$^\vee$) Dual forward cascade:** DNS supports positive mean energy transfer at $R_\lambda \gtrsim 100$. Backscatter (localised inverse transfer) occurs, so pointwise $\Pi_j \ge \Pi_0$ is too strong as a primitive condition; the relevant test is the weighted Abel pairing (13) [10, 9, 14, 15].

- **(DFC2) Spectral steepness:** The compensated spectrum $E(k)\,k^{5/3}/\varepsilon^{2/3}$ is approximately constant ($\approx C_K$) in the inertial range and *decreasing* beyond the bottleneck. At $R_\lambda \gtrsim 200$, the ratio E_j/E_j^* is non-increasing across the inertial range in all published DNS [9, 10]. Deviations from monotonicity are confined to ~ 2 shells near the dissipation cutoff (bottleneck effect), which can be absorbed into the boundary constant C_{bdry}.

- **(DFC3) Remainder control:** The inequality (14) is a conditional quantitative commutator-localisation hypothesis. It follows from explicit commutator estimates such as (23) only when an appropriate Besov/localisation bound is available, together with DFC2 monotonicity. No uniform Leray–Hopf bound of this kind follows from the energy inequality alone.

Dependency table.

Label	Statement	Type	Source
DFC1$^\vee$	Dual forward cascade	Empirical/projection bridge	[10]
DFC2	Spectral steepness	Empirical (DNS)	[9]
DFC3	Conditional commutator localisation	Assumption/conditional estimate	[5]
Lem. 3.18	DFC $\Rightarrow$ NL$'$	**Theorem**	This paper
Prop. 3.12	NL$'$ $\Rightarrow$ Thm 3.25	**Theorem**	This paper

The logical chain is: DFC1$^\vee$ + DFC2 + DFC3 $\xRightarrow{\text{Lem. 3.18}}$ NL$'$ $\xRightarrow{\text{Prop. 3.12}}$ Thm 3.25. The non-rigorous inputs are DFC1$^\vee$ and DFC2, stated as falsifiable inequalities on the measured weighted spectral flux (13) and compensated slope ($\varphi_{j+1} \le \varphi_j$). The implication DFC $\Rightarrow$ NL$'$ $\Rightarrow$ Thm 3.25 is purely analytic once the dual flux condition is assumed. A reviewer can verify or falsify DFC1$^\vee$–DFC2 from DNS data independently of the mathematical framework.

Remark 3.22 (Independent support for dual DFC1 from vanishing viscosity theory)**.** The dual forward-cascade assumption (DFC1$^\vee$) receives independent theoretical support from the vanishing viscosity programme of Drivas [19]: for any space-time L^3 weak solution of the Euler equations arising as a strong limit of $d \ge 3$-dimensional Navier–Stokes solutions, the Duchon–Robert defect measure $D(u)$ is non-negative, and the Lagrangian forward/backward dispersion measure matches the energy defect in the sense of distributions. The resulting Lagrangian time-asymmetry (particles disperse faster backward than forward in $d \ge 3$) is a rigorous characterisation of the forward cascade direction.

De Rosa, Drivas and Inversi [20] sharpen this further: any bounded Euler solution arising as a zero-viscosity limit of Navier–Stokes solutions cannot have anomalous dissipation supported on a set of Hausdorff dimension smaller than d. This bound is sharp.

These results do *not* prove the Littlewood–Paley projection bridge needed for DFC1$^\vee$, but they provide a mathematically rigorous justification for the physical expectation $D(u) \ge 0$ that underlies the weighted forward-flux pairing. The quantitative weighted inequality remains an empirical/projection input.

The following two lemmas provide the analytic backbone. The first isolates the commutator remainder bound as a self-contained estimate; the second gives the flux decomposition that drives the entire argument.

Lemma 3.23 (Commutator remainder bound)**.** *Let u be a Leray–Hopf weak solution on $\mathbb{T}^3$ and let R_j be the Littlewood–Paley commutator remainder from the decomposition $T_j = \Pi_{j-1} - \Pi_j + R_j$.*

(i) $\sum_j R_j = 0$ *(conservation).*

(ii) *If $u \in L^3_t B^{1/3}_{3,\infty}(\mathbb{T}^3)$ with norm $\le B$, then for any sub-window $[j_1, j_2]$ and any bounded weights $(\varphi_j)_j$:*

$$\sum_{j=j_1}^{j_2} |\varphi_j|\,|R_j| \;\le\; C_0\, B^3 \sup_{j\in[j_1,j_2]} |\varphi_j|, \tag{21}$$

where $C_0 > 0$ is a universal constant.

Proof. Part (i) follows from $\sum_j T_j = 0$ and $\sum_j(\Pi_{j-1} - \Pi_j) = 0$ (telescoping). Part (ii) is the standard paraproduct/commutator estimate: each R_j collects terms of the form

$[\Delta_j, S_{j'}](u\otimes u)$, and the off-diagonal decay $\|[\Delta_j, S_{j'}]\|_{L^1\to L^1} \le C\,2^{-|j-j'|/3}$ combined with summing over j' gives $\|R_j\|_{L^1} \le C_0\,B^3$; see [5], Proposition 1 and [6], Lemma 2.1. The weighted sum follows from $\sum|\varphi_j|\,|R_j| \le \sup|\varphi_j|\,\sum|R_j| \le \sup|\varphi_j|\cdot C_0 B^3$. No claim is made that every Leray–Hopf solution has a ν-uniform $L^3_t B^{1/3}_{3,\infty}$ bound; when such a bound is available on a window, the estimate above applies. □

The following lemma, based on the distributional energy balance of Duchon and Robert [6], provides the flux decomposition.

Lemma 3.24 (Flux representation)**.**
For any family of weights $(\varphi_j)_j$ with $\sup_j|\varphi_j| < \infty$, the nonlinear transfer admits the decomposition

$$\sum_j \varphi_j\,T_j \;=\; \sum_j(\varphi_{j+1}-\varphi_j)\,\Pi_j \;+\; \sum_j \varphi_j\,R_j, \tag{22}$$

where Π_j is the energy flux through shell j (eq. (5)) and the commutator remainder R_j collects the Littlewood–Paley paraproduct commutator terms; $\sum_j R_j = 0$. For any sub-window $[j_1, j_2]$, the weighted remainder satisfies the explicit bound

$$\sum_{j=j_1}^{j_2} |\varphi_j\,R_j| \;\le\; C_{\mathrm{rem}}\,\|u\|^3_{L^3_t B^{1/3}_{3,\infty}} \;\cdot\; \sup_{j\in[j_1,j_2]}|\varphi_j|, \tag{23}$$

where $C_{\mathrm{rem}} > 0$ is a universal constant (independent of the window, the solution, and the reference spectrum). This follows from the standard paraproduct decomposition $R_j = \sum_\ell [T_\ell, \Delta_j]$ and the commutator estimate $\|[T_\ell, \Delta_j]\|_{L^1} \le C\,2^{-|j-\ell|/3}\,\|u\|^3_{B^{1/3}_{3,\infty}}$ (see [5], Proposition 1; [6], Lemma 2.1).

Proof (sketch). Write $T_j = \Pi_{j-1} - \Pi_j + R_j$, where $\Pi_j = -\langle \Delta_{\le j}u\otimes\Delta_{\le j}u,\, \nabla\Delta_{\le j}u\rangle$ is the cumulative flux and R_j collects the Littlewood–Paley commutator terms. Abel summation gives (22). The commutator estimate follows from the Duchon–Robert distributional energy balance and standard paraproduct estimates in Besov spaces; see [6] and [5]. □

Theorem 3.25 (Conditional cascade compatibility and energy-input lower bound)**.**
Let $(u^\nu)_{\nu>0}$ be a family of Leray–Hopf weak solutions of the incompressible Navier–Stokes equations on $\mathbb{T}^3$:

$$\partial_t u^\nu + (u^\nu\cdot\nabla)\,u^\nu + \nabla p^\nu = \nu\,\Delta u^\nu + f, \qquad \nabla\cdot u^\nu = 0, \tag{24}$$

with a fixed smooth forcing f and initial data satisfying $\|u_0^\nu\|_{L^2} \le M$. Denote the time-averaged dissipation rate

$$\varepsilon_\nu \;:=\; \frac{\nu}{T}\int_0^T\!\!\int_{\mathbb{T}^3} |\nabla u^\nu|^2\,dx\,dt. \tag{25}$$

Assume that the cascade satisfies one *of the following (in decreasing order of strength):*

(H1) *Entropy compatibility* (11) *(Definition 3.9), or*

(H2) *Window-entropy compatibility* (12) *(Definition 3.11), or*

(H3) *The Downhill Flux Condition* (DFC1$^\vee$), (DFC2), (DFC3) *of Definition 3.14.*

Further assume (ν-uniform energy injection)*: there exists* $m_0 > 0$*, independent of* ν*, such that* $T^{-1}\int_0^T \langle f, u^\nu\rangle\, dt \geq m_0$ *for all* $\nu > 0$*.*

Then the cascade assumptions provide the entropy-compatibility route used in the variational framework. Moreover, by the global energy balance and the uniform energy-injection assumption, there exists a constant $\varepsilon_* > 0$*, depending only on* f *and* M*, such that*

$$\liminf_{\nu\to 0}\ \varepsilon_\nu \ \geq\ \varepsilon_*. \tag{26}$$

Proof (sketch). **Step 1: Energy balance.** Multiplying (24) by u^ν and integrating yields the global energy balance

$$\frac{d}{dt}\frac{1}{2}\|u^\nu\|_{L^2}^2 \ =\ -\nu\|\nabla u^\nu\|_{L^2}^2 \ +\ \langle f, u^\nu\rangle. \tag{27}$$

In the statistically stationary state, the time-averaged dissipation equals the time-averaged energy input: $\varepsilon_\nu = T^{-1}\int_0^T \langle f, u^\nu\rangle\, dt$.

Step 2: Regularised free-energy monotonicity. For $\delta > 0$, define the regularised free energy

$$\mathcal{F}_\delta[\mathbf{E}] \ :=\ \sum_j \Big[(E_j+\delta)\,\ln\frac{E_j+\delta}{E_j^*+\delta}\ -\ (E_j - E_j^*)\Big].$$

This is C^1 in all arguments and satisfies $\mathcal{F}_\delta \geq 0$ with equality iff $\mathbf{E} = \mathbf{E}^*$. Computing $\frac{d}{dt}\mathcal{F}_\delta$ along the shell-energy evolution $\frac{1}{2}\frac{d}{dt}E_j = -\nu k_j^2 E_j + T_j + \langle \Delta_j u, \Delta_j f\rangle$ (up to commutator corrections), the three contributions are:

1. *Viscous term.* It contributes

$$-2\nu\sum_j k_j^2\, E_j^\nu\ \ln\!\Big(\tfrac{E_j^\nu+\delta}{E_j^*+\delta}\Big).$$

   Individual summands $k_j^2\, E_j\ \ln(E_j/E_j^*)$ can have either sign: they are positive when $E_j > E_j^*$ and negative when $E_j < E_j^*$. Hence the viscous contribution has no fixed sign. Its role in (28) is instead dissipative: it pushes the spectrum toward $\mathbf{E}^*$, because shells with $E_j > E_j^*$ are damped faster than equilibrium, while shells with $E_j < E_j^*$ are damped more slowly;

2. *Nonlinear transfer.* By Assumption (11) (entropy-compatible cascade),

$$\sum_j \ln\!\Big(\tfrac{E_j+\delta}{E_j^*+\delta}\Big)\, T_j \leq 0;$$

3. *Forcing:* bounded by $C\,\|f\|_{H^s}^2$ via Cauchy–Schwarz.

Combining these and passing $\delta \downarrow 0$ via Fatou's lemma (lower semicontinuity of $\mathcal{F}$) yields

$$\frac{d}{dt}\,\mathcal{F}[\mathbf{E}^\nu(t)] \ \leq\ -2\nu\sum_j k_j^2\, E_j^\nu\ \ln\frac{E_j^\nu}{E_j^*}\ +\ C\,\|f\|_{H^s}^2. \tag{28}$$

The quantity $D_{\mathcal{F}}^\nu := \sum_j k_j^2\, E_j^\nu\ \ln(E_j^\nu/E_j^*)$ is the *entropy dissipation*. It is not sign-definite (individual terms can be positive or negative). Each summand $k_j^2\, E_j\ \ln(E_j/E_j^*)$ vanishes if and only if $E_j = E_j^*$ (since $\ln(x/y) = 0$ iff $x = y$ for $x, y > 0$, and $k_j^2\, E_j > 0$); hence $D_{\mathcal{F}}^\nu = 0$ with all summands individually zero if and only if $\mathbf{E}^\nu = \mathbf{E}^*$. The dissipative character of the viscous term is encoded in the full inequality (28): combined with the non-positive nonlinear contribution and the bounded forcing, it ensures that $\mathcal{F}$ decreases on average.

Step 3: Lower bound on dissipation (using ND). Under the ν-uniform energy injection assumption (stated in Step 4 below), the energy balance (Step 1) yields $\varepsilon_\nu \geq m_0 > 0$ directly. Without this assumption, one can only argue heuristically: if $\varepsilon_\nu \to 0$, then for smooth, non-trivial f and bounded $\|u^\nu\|_{L^2} \leq C(M)$, the forcing term $\langle f, u^\nu \rangle$ remains bounded but the dissipation vanishes, which is inconsistent with sustained energy injection. The ND assumption formalises this.

Step 4: Quantitative bound. The $\mathcal{F}$-monotonicity (28) controls the entropy dissipation $D^\nu_{\mathcal{F}}(t) := \sum_j k_j^2 \, E_j^\nu \, \ln(E_j^\nu / E_j^*)$, which measures the spectral distance from the Kolmogorov reference $\mathbf{E}^*$. However, the physical dissipation $\varepsilon_\nu = \nu \sum_j k_j^2 E_j^\nu$ is not directly controlled by $D^\nu_{\mathcal{F}}$ without an additional *non-degeneracy* assumption linking these two quantities.

Assumption ND (ν-uniform energy injection). There exists $m_0 > 0$, independent of ν, such that

$$\frac{1}{T} \int_0^T \langle f, u^\nu \rangle \, dt \;\geq\; m_0 > 0 \qquad \text{for all } \nu > 0. \tag{29}$$

Under this assumption, the energy balance (Step 1) immediately gives $\varepsilon_\nu \geq m_0$ for all ν, hence $\liminf_{\nu \to 0} \varepsilon_\nu \geq m_0 > 0$.

A refined bound can be obtained as follows. For divergence-free $f \in H^{-1}(\mathbb{T}^3)$ with $\langle f, \cdot \rangle$ non-trivial on divergence-free fields, the Poincaré inequality on $\mathbb{T}^3$ gives $\|u^\nu\|_{L^2} \leq C_P \, \|\nabla u^\nu\|_{L^2}$, so that $\langle f, u^\nu \rangle \leq \|f\|_{H^{-1}} \, \|\nabla u^\nu\|_{L^2} \leq \|f\|_{H^{-1}} \, (\varepsilon_\nu/\nu)^{1/2}$. Combining with (29): $m_0 \leq \|f\|_{H^{-1}} \, (\varepsilon_\nu/\nu)^{1/2}$, which gives $\varepsilon_\nu \geq \nu \, m_0^2 / \|f\|^2_{H^{-1}}$. This bound is non-trivial only for fixed ν and degenerates as $\nu \to 0$. The *ν-independent* lower bound $\varepsilon_* \geq m_0$ from Step 1 remains the operative estimate; a sharper ν-independent bound of the form $\varepsilon_* \geq c \, \|f\|^2_{H^{-1}} / M^2$ would require a spectral Poincaré inequality linking $D^\nu_{\mathcal{F}}$ to ε_ν, which is an open problem (see the Remark below). The natural Sobolev norm for the energy injection is H^{-1} (dual to $\|\nabla u\|_{L^2}$); the H^s with $s > -1$ appearing in (28) arises from the shell-wise forcing decomposition and is strictly weaker.

□

Remark 3.26 (Role of the free-energy machinery – honest assessment)**.**
Theorem 3.25 contains two logically independent results that should be distinguished:

(a) **Quantitative bound (ND alone).** The bound $\varepsilon_* \geq m_0$ follows directly from the energy balance and assumption ND, *without* invoking the functional $\mathcal{F}$ or hypotheses (H1)–(H3). This is the operative estimate.

(b) **Structural attractor property ($\mathcal{F}$-machinery).** The $\mathcal{F}$-monotonicity (28) provides a *qualitative* explanation for why the K41 profile is the attracting state of the cascade, and it controls the entropy dissipation $D^\nu_{\mathcal{F}}$, which measures spectral distance from K41. This structural insight requires (H1)–(H3) but does *not* yield a sharper quantitative bound than (a) without an additional spectral Poincaré inequality (Conjecture 4.7).

The hypotheses (H1)–(H3) are therefore logically necessary only for the structural attractor interpretation (b), not for the quantitative anomalous-dissipation bound (a). A full quantitative upgrade of (b) remains an open problem.

Remark 3.27 (Scope and validity of the entropy-compatibility hypothesis).
Theorem 3.25 is conditional on Assumption (11). Three settings in which this hypothesis is known or expected to hold:

1. *Shell models:* In GOY and Sabra shell models with a Markov-type transfer structure, the nonlinear term is a genuine mass-preserving operator on shell energies, and (11) follows from the standard entropy-dissipation inequality for relative entropies under doubly stochastic maps.
2. *Stochastic cascade models:* In multiplicative cascade models of turbulence (Mandelbrot, Kahane), the energy redistribution is by construction a Markov kernel, yielding (11).
3. *Navier–Stokes with downhill flux:* For NS solutions whose energy flux Π_j satisfies a "downhill" condition—energy flows preferentially from overexcited to underexcited shells relative to $\mathbf{E}^*$—Lemma 3.24 combined with Abel summation gives (11) up to controllable commutator remainders. Whether this downhill property holds generically for NS solutions in the inertial range is an open question closely related to the Onsager conjecture and the structure of the Duchon–Robert defect measure [6].

4 Intermittency Corrections

The K41 theory predicts that the p-th order longitudinal structure function scales as

$$S_p(\ell) \;:=\; \left\langle |\delta_\ell u|^p \right\rangle \;\sim\; (\varepsilon\,\ell)^{p/3}, \qquad \zeta_p^{\mathrm{K41}} = \frac{p}{3}. \tag{30}$$

Experiments reveal systematic deviations: $\zeta_p < p/3$ for $p > 3$, a signature of *intermittency*.

4.1 Fluctuations around the variational minimum

In our framework, intermittency arises from fluctuations of the shell energies $\mathbf{E}$ around the K41 minimiser $\mathbf{E}^*$. Expanding $\mathcal{F}$ to second order:

$$\mathcal{F}[\mathbf{E}^* + \delta\mathbf{E}] \;=\; \mathcal{F}[\mathbf{E}^*] \;+\; \frac{1}{2}\sum_j \frac{(\delta E_j)^2}{E_j^*} \;+\; O(|\delta\mathbf{E}|^3). \tag{31}$$

The Hessian $H_{jj} = 1/E_j^*$ implies that fluctuations at small scales (large j, small E_j^*) are suppressed more strongly than fluctuations at large scales. This asymmetry is the variational origin of intermittency.

4.2 Connection to the K62 log-normal model

If we model the fluctuations δE_j as Gaussian in the variable $\ln(E_j/E_j^*)$, and further assume that phase correlations between Littlewood–Paley blocks are negligible (so that shell-energy fluctuations translate directly into structure-function scaling), the resulting structure function exponents are

$$\zeta_p \;=\; \frac{p}{3} \;-\; \frac{\mu}{18}\,p(p-3), \tag{32}$$

which is exactly the K62 (Kolmogorov–Obukhov) refined similarity hypothesis with intermittency parameter μ [3, 10]. The parameter μ is determined by the variance of $\ln(E_j/E_j^*)$ under the Gibbs measure $\exp(-\mathcal{F}/\Theta)$, where the effective temperature $\Theta > 0$ parametrises the strength of turbulent fluctuations around the K41 equilibrium (analogous to the temperature in a canonical ensemble). In the limit $\Theta \to 0$, the Gibbs measure concentrates on the K41 minimiser and $\mu \to 0$ (no intermittency); for finite Θ, the log-normal variance scales as $\mu \propto \Theta\, E_j^*$, connecting the intermittency parameter to the intensity of the cascade fluctuations.

Remark 4.1.
The She–Lévêque model [8] proposes the alternative

$$\zeta_p = \frac{p}{9} + 2\Big(1 - \left(\tfrac{2}{3}\right)^{p/3}\Big),$$

which fits experimental data more accurately for high p. Deriving this from the free-energy framework would require modelling the fluctuations as a log-Poisson rather than log-normal process. This is an open problem within our approach.

4.3 Structural Classification

The joint minimisation of Theorem 3.4 exhibits a three-level hierarchy: (i) the leading-order energy budget is neutral (any power-law spectrum is kinematically admissible); (ii) the first-order constant-flux constraint restricts but does not uniquely determine the profile; (iii) the second-order strict convexity of $\mathcal{F}$ selects K41 as the unique minimiser. This pattern—referred to as *second-order resolvent dominance* in [17]—has analogues in other variational selection problems and is discussed in detail there.

4.4 Robustness guardrails and formal penalty representation

The strict-DFC framework of Definition 3.14 requires the dual weighted lower bound (13) and exact monotonicity of the compensated spectrum. DNS data exhibit small-scale fluctuations around these idealisations, but two previously tempting extensions need to be kept as guardrails rather than new unconditional theorems.

Proposition 4.2 (Relaxed flux constraint: scope caveat)**.**
Let $\mathcal{A}_{\text{slack}}$ be any relaxed admissible class that contains the exact K41 profile $\mathbf{E}^*$ and keeps the objective equal to $\mathcal{F}[\mathbf{E}]$. Then minimising $\mathcal{F}$ over $\mathcal{A}_{\text{slack}}$ still gives $\mathbf{E}^*$ exactly. A non-trivial slack theorem requires an additional data-fit term, a perturbed reference, or a constraint that excludes $\mathbf{E}^*$.

Proof. By construction, $\mathcal{F}[\mathbf{E}] \geq 0$ and $\mathcal{F}[\mathbf{E}^*] = 0$. If $\mathbf{E}^* \in \mathcal{A}_{\text{slack}}$, no feasible profile can have lower objective value. Hence the relaxed problem is exact rather than a genuine robustness estimate. □

Remark 4.3.
Finite-Reynolds-number robustness should therefore be formulated as a statistical or inverse problem: measured spectra are compared against the K41 reference with explicit data errors and projection errors. The relaxed DFC inequalities are useful diagnostics, not by themselves a displacement bound away from $\mathbf{E}^*$.

Proposition 4.4 (Formal penalty representation)**.**
In a finite shell truncation, the pair $(\mathbf{E}^*, \mathbf{\Pi}^*)$ *with* $\Pi_j^* = \varepsilon$ *is a saddle point of the decoupled penalty functional*

$$\mathcal{L}(\mathbf{E}, \mathbf{\Pi}) = \mathcal{F}[\mathbf{E}] - \lambda \sum_j (\Pi_j - \varepsilon)^2, \qquad \lambda > 0, \tag{33}$$

on the product of positive spectra and bounded flux profiles. This is a formal penalty representation of K41 plus constant flux, not a dynamical two-player theorem.

Proof. The $\mathbf{E}$ and $\mathbf{\Pi}$ variables are decoupled in (33). The unique minimiser of $\mathcal{F}$ is $\mathbf{E}^*$, and the unique maximiser of the negative quadratic penalty is $\Pi_j = \varepsilon$ for every shell. The stated saddle inequalities therefore follow directly. □

Remark 4.5.
The minimax notation is useful bookkeeping for the constant-flux penalty. A genuine game-theoretic claim would require a coupled payoff in which the flux variables are generated by the nonlinear dynamics and react to the chosen spectrum.

Remark 4.6 (Weighted ℓ^2-stability from the Hessian bound)**.**
The functional $\mathcal{F}[E]$ is uniformly convex about the K41 minimiser E^* with diagonal Hessian

$$D^2\mathcal{F}\Big|_{E^*} = \mathrm{diag}(1/E_j^*).$$

A direct Taylor expansion, without appeal to optimal transport theory, yields the weighted stability bound

$$\sum_j \frac{(E_j - E_j^*)^2}{E_j^*} \;\le\; 2\,\mathcal{F}[E]$$

for all admissible spectra E with $E_j > 0$. This provides a Reynolds-number-independent stability guarantee: perturbations of the K41 profile are controlled in the $\ell^2(1/E_j^*)$-norm by the free-energy excess. The bound complements the DFC-slack estimate (Proposition 4.2) and follows from the elementary inequality

$$x\ln(x/y) - x + y \;\ge\; \frac{(x-y)^2}{2y} \qquad (x, y > 0).$$

4.5 Open conjectures and intrinsic characterisation

The variational framework developed above suggests several precise conjectures that connect the free-energy structure to classical turbulence phenomenology.

Conjecture 4.7 (Spectral Poincaré inequality)**.**
For the energy spectrum $E(k)$ *of a statistically stationary turbulent flow at viscosity* ν*, the entropy dissipation* $D_F^\nu = -dF[E]/dt$ *and the physical dissipation* $\varepsilon_\nu = 2\nu \int k^2 E(k)\, dk$ *are related by a spectral Poincaré inequality:*

$$D_F^\nu \ge \kappa_\nu \cdot \varepsilon_\nu,$$

where $\kappa_\nu > 0$ *is the spectral Poincaré constant. If* κ_ν *remains bounded below as* $\nu \to 0$ *(i.e.,* $\kappa_\nu \ge \kappa_0 > 0$*), then the K41 minimiser is dynamically stable in the inviscid limit: perturbations of* E^* *are dissipated at a rate proportional to the physical dissipation.*

Remark 4.8.
The spectral Poincaré inequality bridges the thermodynamic (entropy-based) and physical (energy-based) descriptions of turbulence. Its validity would imply that the K41 spectrum is not merely a static minimiser but an attractor of the turbulent dynamics.

Threshold axiom and diagnostic framework

The spectral Poincaré inequality admits a precise formulation as a *threshold axiom*—the minimal structural condition that separates dissipative cascade dynamics from anomalous trapping.

Axiom 4.9 (Spectral Poincaré – SP-TU)**.**
Let $\mathcal{H} = \{E = (E_j)_{j\geq 1} : E_j \geq 0, \sum E_j < \infty\}$ be the space of shell energy distributions, let $\mathcal{N} \subset \mathcal{H}$ be the subspace of null modes (conservation laws: total energy $\sum E_j = \mathrm{const}$), and let $\mu_{\mathcal{F}}$ be the Gibbs measure $\mu_{\mathcal{F}} \propto e^{-\mathcal{F}[E]/\Theta}$. There exists a spectral gap $\lambda > 0$ such that

$$\mathrm{Var}_{\mu_{\mathcal{F}}}(g) \leq \frac{1}{\lambda}\, \mathcal{E}_{\mathcal{F}}(g, g) \qquad \textit{for all } g \perp \mathcal{N}, \tag{34}$$

where $\mathcal{E}_{\mathcal{F}}(g, g) = \int |\nabla g|^2 \, d\mu_{\mathcal{F}}$ is the Dirichlet form associated with the free-energy landscape.

Proposition 4.10 (Equivalences of SP-TU)**.**
The following are equivalent:

(i) ***Spectral gap:*** *Axiom 4.9 holds with $\lambda > 0$.*

(ii) ***Exponential relaxation:*** *For the semigroup P_t of the energy cascade dynamics, $\mathrm{Var}_{\mu_{\mathcal{F}}}(P_t g) \leq e^{-2\lambda t}\, \mathrm{Var}_{\mu_{\mathcal{F}}}(g)$ for all $g \perp \mathcal{N}$.*

(iii) ***Rayleigh characterisation:*** *$\lambda = \inf_{g\perp\mathcal{N},\, g\neq 0} \frac{\mathcal{E}_{\mathcal{F}}(g,g)}{\mathrm{Var}_{\mu_{\mathcal{F}}}(g)}$.*

Proof sketch. (i)⇔(iii) is the standard variational characterisation of the spectral gap. (i)⇒(ii) follows from the spectral theorem for self-adjoint generators; (ii)⇒(i) by differentiating at $t = 0$. □

Remark 4.11 (Diagnostic interface)**.**
The Rayleigh characterisation (iii) provides both analytical and numerical access to λ:

- **Analytical:** Test functions of the form $g_j = \ln(E_j/E_j^*)$ yield $\mathcal{E}_{\mathcal{F}}(g_j, g_j)/\mathrm{Var}(g_j) = 1/E_j^*$, so $\lambda \geq \min_j(1/E_j^*) = 1/(C_K\varepsilon^{2/3}k_1^{-5/3}\Delta k_1) > 0$. This is the diagonal Hessian bound (Remark 3.5).
- **Numerical:** Estimate λ from the exponential decay of autocorrelations $\langle g(t)\, g(0)\rangle_{\mu_{\mathcal{F}}} \sim e^{-\lambda t}$ in shell-model simulations (cf. Remark 5.3).
- **Block decomposition:** Establish local Poincaré on shell clusters $[j_1, j_2]$ and glue via controlled overlap (Zegarlinski block dynamics, analogous to the Yang–Mills strategy in the companion paper [22]).

Remark 4.12 (No-go mechanisms)**.**
SP-TU fails if and only if there exist *quasi-null modes*: a sequence $g_k \perp \mathcal{N}$ with $\mathrm{Var}(g_k) = 1$ and $\mathcal{E}(g_k, g_k) \to 0$. Three physical mechanisms can produce such modes:

1. **Metastability:** The cascade dynamics traps in nearly invariant regions (e.g., bottleneck accumulation), creating slow indicator observables with vanishing Dirichlet form.

2. **Multi-scale leakage:** Dissipation acts on small scales while variance "parks" on large scales, producing a defective coercivity that cannot be closed by a single λ.

3. **Intermittency bursts:** Rare, intense events create heavy-tailed fluctuations that violate the Gaussian assumption underlying the Hessian bound.

For K41 turbulence in the inertial range, none of these mechanisms is expected to dominate: the dual forward cascade (DFC1$^\vee$) suppresses trapping in the weighted Abel direction, the scale locality (A1) prevents leakage, and the log-normal intermittency (Section 4) decays as $1/N_{\text{shells}}$ (Remark 4.18). SP-TU is therefore *expected* to hold, but a rigorous proof remains open.

Lemma 4.13 (SP-TU implies universal dissipation control)**.**
If Axiom 4.9 holds, then for any perturbation $\delta E = E - E^*$ with $\sum \delta E_j = 0$:

$$\sum_j |\delta E_j|^2/E_j^* \leq \frac{1}{\lambda}\,\mathcal{F}[E]. \tag{35}$$

In particular, the sublevel sets $\{E : \mathcal{F}[E] \leq C\}$ are precompact in ℓ^1, providing the DS3 condition for the Dissipative Selection Principle.

Proof sketch. By the Poincaré inequality (34) applied to $g = \delta E/E^*$, the $L^2(\mu_{\mathcal{F}})$-norm of g is controlled by the Dirichlet form, which equals $\mathcal{F}[E]$ to leading order (Hessian approximation). The precompactness follows since the weighted ℓ^2-bound with weights $1/E_j^* \sim k_j^{2/3}$ controls the tail of the energy distribution. □

Remark 4.14 (Finite-N spectral gap via shell-model Markov structure)**.**
For finite-dimensional truncations, namely GOY/SABRA shell models with N shells plus stochastic forcing, the shell-to-shell energy transfer matrix T_{ij} can be normalised to a stochastic kernel. If this kernel is irreducible and satisfies the Dobrushin-type condition

$$\eta := \max_i \sum_{j\neq i} |T_{ij} - T'_{ij}| < 1,$$

which holds for sufficiently strong forcing relative to the nonlinearity, then standard Markov-chain theory yields a Poincaré inequality with gap

$$\Delta_{\text{spectral}} \geq 1 - \eta > 0.$$

This provides a rigorous finite-N analogue of the conjectured spectral Poincaré inequality and establishes a direct structural parallel with the Yang–Mills Dobrushin–Zegarlinski machinery [22]. Extending this argument to the limit $N \to \infty$ requires control of $\eta(N)$, precisely the turbulence analogue of the Yang–Mills continuum limit.

Remark 4.15 (Intrinsic characterisation of E^* via Kármán–Howarth–Monin)**.**
The Kármán–Howarth–Monin relation

$$\partial_t\langle|\delta u|^2\rangle + \nabla_r \cdot \langle|\delta u|^2\delta u\rangle = -4\varepsilon/3 + 2\nu\Delta_r\langle|\delta u|^2\rangle$$

provides an intrinsic constant-flux anchor without assuming K41 *a priori*. In the inertial range ($\eta \ll r \ll L$), the viscous term vanishes and the stationary KHM relation yields

$$\langle |\delta u|^2 \delta u \rangle = -4\varepsilon r/5,$$

that is, the four-fifths law. This supports the flux input used by the variational scheme, but it does not by Fourier transformation alone determine the second-order spectrum uniquely. Recovering $E(k) \sim \varepsilon^{2/3} k^{-5/3}$ still requires the usual self-similarity, locality, or closure input. Thus KHM is a consistency check on the reference spectrum, not an independent uniqueness proof.

Conjecture 4.16 (Intermittency exponents from Hessian fluctuations)**.**
The anomalous scaling exponents ζ_p of the structure functions $S_p(r) = \langle |\delta u(r)|^p \rangle \sim r^{\zeta_p}$ can be expressed in terms of the Hessian of the free energy functional $F[E]$ at the K41 minimiser:

$$\zeta_p = \frac{p}{3} - \frac{p(p-3)}{2} \sigma_H^2 + O(\sigma_H^4),$$

where $\sigma_H^2 = \mathrm{tr}(\nabla^2 F[E^*]^{-1})/N_{\text{shells}}$ is the normalised inverse Hessian trace ("fluctuation strength"). For $\sigma_H^2 \to 0$ (infinite Reynolds number, Hessian dominance), $\zeta_p \to p/3$ (K41). For finite $\sigma_H^2 > 0$, the corrections reproduce the She–Lévêque hierarchy $\zeta_p = p/9 + 2(1 - (2/3)^{p/3})$ if $\sigma_H^2 = 2\ln(3/2)/9 \approx 0.090$.

Remark 4.17.
This conjecture connects the variational framework to the phenomenology of intermittency: the curvature of $F[E]$ at the minimiser controls not only stability (Proposition 4.2) but also the statistics of fluctuations around it.

Remark 4.18 (Intermittency parameter disambiguation)**.**
The Hessian-trace prediction $\sigma_H^2 \sim 1/N_{\text{shells}}$ quantifies the anomalous correction $\Delta\zeta_2 = \zeta_2 - 2/3$ to the energy spectrum exponent, not the Kolmogorov–Obukhov intermittency parameter μ (which satisfies $\mu \approx 0.2$–0.3). The She–Lévêque model (1994) predicts $\zeta_p = p/9 + 2(1 - (2/3)^{p/3})$, yielding $\Delta\zeta_2 \approx 0.03$ for $N \approx 30$ resolved shells, consistent with our $1/N$ prediction.

Result classification

Result	Type	Reference
Theorems (rigorous):		
$\mathcal{F}[E] \geq 0$, $= 0$ iff $E = E^*$	Theorem	Thm. 3.4
DFC$^\vee$ $\Rightarrow$ NL'	Lemma	Lem. 3.18
Energy input $\Rightarrow$ dissipation lower bound	Theorem	Thm. 3.25
Local slope condition implies DFC2	Proposition	Prop. 3.15
Empirical hypotheses (DNS-verifiable):		
DFC1$^\vee$ (dual forward cascade)	Empirical/projection bridge	Def. 3.14, $R_\lambda \gtrsim 100$
Numerically confirmed:		
$\mathcal{F}[E_{K41}] = 0$ (minimiser)	Exact	`compute_F_spectrum.py`
DFC2 (Sabra shell model)	Shell model	Rem. 5.3
$\mathcal{F} > 0$ for 500/500 snapshots	Shell model	Rem. 5.3
Initial DFC1$^\vee$ ledger controls	Ledger/toy controls	Table 1
DNS-falsifiable predictions:		
$\mathcal{F}[E_{\mathrm{DNS}}] \to 0$ as $R_\lambda \to \infty$	Prediction	Sec. 5
φ_j non-increasing in inertial range	Prediction	DFC2

5 Discussion

The joint minimisation over all admissible closures is not a tautological restatement of K41: Kolmogorov's original derivation relies on dimensional analysis, while our Theorem shows K41 is the unique minimiser of a well-defined variational problem, providing a structural explanation for universality.

5.1 Relation to the Onsager conjecture

Theorem 3.25 separates two issues. The lower bound on dissipation follows from the global energy balance plus the stated ν-uniform energy-input assumption. The variational contribution is the conditional route from DFC1$^\vee$–DFC2–DFC3 to entropy-compatible cascade structure and K41 selection. This is complementary to the convex-integration programme [7, 26, 21], but it is not an unconditional selection theorem for all Navier–Stokes weak solutions.

5.2 The role of the closure family

The reformulation of Theorem 3.4 via the admissible flux family (Definition 3.1) significantly weakens the dependence on any specific closure model. The K41 spectrum is selected not by a single algebraic relation but as the unique minimiser across *all* local, dimensionally consistent, monotone flux operators. Nevertheless, axiom (A2) still imposes a structural constraint—the $k_j\, E_j^{3/2}$ scaling prefactor—that ultimately encodes the Kolmogorov dimensional argument. Deriving this prefactor from the Kármán–Howarth–Monin relation or from rigorous bounds on the Littlewood–Paley trilinear form remains an important open problem.

5.3 Regime selection and the laminar-turbulent transition

The variational principle of Theorem 3.4 selects the K41 spectrum *within* the turbulent regime, but does not explain the laminar-turbulent transition itself. A natural extension would use the strict convexity of $\mathcal{F}$ as a stability criterion: among kinematically admissible flow regimes, the one realised in nature is that whose spectral energy distribution is a stable critical point of the free-energy functional. Exploring this connection quantitatively is left to future work.

5.4 Limitations

We collect the principal limitations of the present work:

1. **Conditionality.** Theorem 3.25 is conditional on an entropy-compatibility hypothesis (or on the sufficient hierarchy DFC1$^\vee$–DFC2–DFC3). DFC1$^\vee$ is motivated by positive coarse-grained flux, but the projection bridge from Duchon–Robert/Eyink scale flux to the weighted Littlewood–Paley pairing is not proved here. The DFC3 condition requires a *quantitative bound* on the commutator remainder (eq. (14)). Such a bound follows in regularity/localisation regimes where the commutator estimate is available, but it is not automatic from the Leray–Hopf energy inequality. Whether a useful DFC3 bound holds for general turbulent vanishing-viscosity families remains open.

2. **Inertial-range idealisation.** The functional $\mathcal{F}$ is defined on the inertial range $\mathcal{J}$, ignoring the forcing and dissipation ranges. Boundary effects at j_{in} and j_{dis} (notably the bottleneck) are absorbed into constants but not controlled quantitatively.

3. **Intermittency.** The K62 connection (Section 4) is qualitative. Rigorous derivation of intermittency exponents from the Hessian fluctuations remains open.

4. **Uniqueness of the reference.** While Remark 3.6 argues that $\mathbf{E}^*$ is the only physically consistent reference, a fully intrinsic characterisation of $\mathbf{E}^*$ (without *a priori* appeal to K41) would strengthen the argument.

5.5 Numerical evidence and proposed DNS test

Direct numerical simulations (DNS) at Taylor-microscale Reynolds numbers $R_\lambda \sim 200$–1000 consistently produce energy spectra close to $k^{-5/3}$ in the inertial range, with a Kolmogorov constant $C_K \approx 1.5$ [9, 10, 13]; high-Reynolds boundary-layer benchmarks are consistent with local-isotropy diagnostics [23]. We propose the following reproducible test protocol:

1. Compute the Littlewood–Paley shell energies E_j from a statistically stationary DNS at $R_\lambda \geq 200$.

2. Evaluate $\mathcal{F}[\mathbf{E}]$ using the measured $\mathbf{E}$ and the K41 reference $\mathbf{E}^*$, where $\varepsilon := \nu \langle \|\nabla u\|_{L^2}^2 \rangle_t$ is the time-averaged viscous dissipation rate from the same DNS run.

3. Verify that $\mathcal{F}$ decreases with increasing R_λ and that the compensated ratio $\varphi_j = \ln(E_j/E_j^*)$ is non-increasing in j across the inertial range (DFC2).
4. Compute the spectral flux Π_j and verify the weighted inequality (13) for $a_j = \varphi_j - \varphi_{j+1} \geq 0$ across inertial windows (DFC1$^\vee$). Pointwise $\Pi_j \geq \Pi_0 > 0$ is a stronger optional test.

Steps 3–4 constitute a direct, falsifiable test of the Downhill Flux Condition that is independent of the mathematical framework.

Implementation notes. (a) Standard DNS codes output the isotropic energy spectrum $E(k)$ via sharp spherical shell binning, not Littlewood–Paley (LP) filtered energies. For smooth spectra, the difference is of order $O(\Delta k_j/k_j) = O(\ln 2)$ per shell and can be estimated by comparing sharp and smooth binning. (b) Steps 1–4 should be applied to instantaneous spectra at multiple statistically independent time snapshots within the stationary regime; ensemble averaging of $\mathcal{F}$ and φ_j then provides confidence intervals for the DFC verification.

5.6 Falsification protocol

The predictions of this framework can be tested against direct numerical simulation (DNS) data as follows:

1. **Spectral data:** Extract the energy spectrum $E(k)$ from a well-resolved DNS at Taylor-microscale Reynolds number $Re_\lambda \geq 200$.
2. **Functional evaluation:** Compute $F[E] = D_{\mathrm{KL}}(E\|E^*) + \lambda \sum_j |\Pi_j[E] - \varepsilon|$ using shell-averaged spectra with logarithmic binning (shell ratio $r = 2^{1/3}$).
3. **Minimiser comparison:** Compare the DNS spectrum $E_{\mathrm{DNS}}(k)$ with the K41 prediction $E^*(k) = C_K \varepsilon^{2/3} k^{-5/3}$ in the inertial range $k_f \ll k \ll k_\eta$.
4. **Falsification criterion.** Since $\mathcal{F}[E^*] = 0$ by construction, the functional cannot be falsified by finding $\mathcal{F}[E_{\mathrm{DNS}}] < 0$. Instead, the DFC route is challenged if DFC1$^\vee$ or DFC2 is *violated* at high R_λ, where the theory predicts they should hold, or if $\mathcal{F}[E_{\mathrm{DNS}}]$ fails to decrease with increasing R_λ, indicating that K41 is not the asymptotic attractor. Pointwise failures of $\Pi_j \geq \Pi_0$ alone do not refute the dual route.
5. **Intermittency test:** Compute the diagonal curvature values $\lambda_j = \partial^2 F/\partial E_j^2 = 1/E_j$ and compare their weighted profile with the K41 reference values $1/E_j^*$. These eigenvalues are positive for every positive spectrum, so negative Hessian eigenvalues of the KL functional are *not* a valid instability diagnostic. The relevant falsification signal is instead failure of the DFC/SP-TU conditions or growth of the spectral distance from E^* with increasing R_λ.

Remark 5.1 (DFC as energetic stability class)**.**
The Downhill Flux Condition should be tested as a dual/windowed energetic stability diagnostic rather than as a pointwise shell axiom. Publicly available DNS datasets (JHTDB [24], $\mathrm{Re}_\lambda \sim 200$) are suitable for this test because it involves relative free-energy differences and Abel-weighted flux pairings rather than asymptotic exponents alone. In the presence of non-unique Leray–Hopf solutions, DFC would provide a falsifiable selection diagnostic; the present paper does not prove that it selects the physically realised branch.

Remark 5.2 (Numerical verification of Theorem 3.4)**.**
A numerical evaluation of $\mathcal{F}[E]$ over six synthetic energy spectra confirms that the K41 spectrum is the unique minimiser: $\mathcal{F}[E_{\mathrm{K41}}] = 0$ exactly, while $\mathcal{F}[E_{\mathrm{K62}}] = 2.0 \times 10^{-4}$ (intermittency correction), $\mathcal{F}[E_{k^{-2}}] = 1.74$ (steep), $\mathcal{F}[E_{k^{-4/3}}] = 5.94$ (shallow), and $\mathcal{F}[E_{\mathrm{thermal}}] = 7.55$ (equipartition). A perturbation test (300 random perturbations δE at three noise levels $\|\delta E\|/\|E^*\| \in \{0.01, 0.1, 1.0\}$) yields $\mathcal{F}[E^* + \delta E] > 0$ for *all* 300 samples, confirming strict convexity numerically. The DFC conditions are satisfied by K41 and K62 (with She–Léveque intermittency $\mu = 0.03$), but violated by steeper, shallower, or thermal spectra. Numerical scripts: `compute_F_spectrum.py`.

Remark 5.3 (Sabra shell model validation of DFC)**.**
Pointwise DFC1 and DFC2 were tested on the Sabra shell model [25] with $N = 22$ shells, $\nu = 10^{-7}$, using an IMEX integrator (500 snapshots). DFC2 (spectral steepness) is strongly confirmed: inertial-range slopes are -0.663, -0.675, -0.661, -0.699, -0.700 for shells $n = 5, \ldots, 9$ (K41 prediction: $-2/3 \approx -0.667$, deviation $< 2\%$). The free energy satisfies $\mathcal{F}[E] > 0$ for all 500 snapshots (100%), with $\mathcal{F}[\langle E \rangle] = 0.198$, confirming the K41 spectrum as approximate minimizer. The pointwise flux Π_n is sign-alternating, a known feature of shell models; this motivates testing DFC1$^\vee$ by the weighted Abel pairing instead of by pointwise positivity. Existing scripts: `compute_goy_shell_dfc.py` and the initial Dual-DFC1 ledger `compute_dual_dfc.py`. The ledger is summarised in Table 1; it keeps the proof status unchanged because DFC1$^\vee$ remains an empirical/projection bridge, but it makes the positive and negative controls explicit.

Scenario	DFC2	DFC1 pt.	DFC1$^\vee$	Ratio	Interpretation
K41 reference	pass	pass	pass	1.000	vacuous K41 reference control
Smooth forward	pass	pass	pass	1.128	smooth forward-cascade positive control
Alternating tolerated	pass	fail	pass	1.020	pointwise backscatter tolerated by the Abel pairing
Bad corridor	pass	fail	fail	0.635	Abel-weighted backscatter corridor rejected
Same-mean shuffle	fail	fail	fail	0.493	same-mean shuffle exposes reference-risk failure

Table 1: Initial Dual-DFC1 ledger produced by `compute_dual_dfc.py`. The ratio column is the Abel-weighted flux divided by the scenario target. The table separates positive controls from backscatter-corridor controls; it is a bookkeeping test, not a DNS proof. Full output: `_results/DUAL_DFC_LEDGER_2026-05-27.md`.

5.7 Extensions

- **Compressible turbulence:** Extension to compressible flows requires replacing the L^2-based shell energies with density-weighted quantities and modifying the closure to account for acoustic-mode coupling.
- **Magnetohydrodynamic (MHD) turbulence:** The Iroshnikov–Kraichnan spectrum $E(k) \sim k^{-3/2}$ should emerge from a modified functional incorporating Alfvén-wave interactions.
- **Two-dimensional turbulence:** The dual cascade (inverse energy cascade with $k^{-5/3}$ and direct enstrophy cascade with k^{-3} [12]) presents an interesting test case

requiring a two-constraint variational problem.

5.8 Post-Zookeeper split: protect Paper A, sharpen Paper B

The Zookeeper transfer clarifies the publication split. Paper A should remain the unconditional variational statement: K41 is the unique minimiser of the scale-resolved free-energy functional under the stated admissibility assumptions. It should not absorb claims about anomalous dissipation, Onsager non-uniqueness, or physical solution selection.

Here too the post-Zookeeper import is RFEP v1.8 normal-form discipline, not a new unconditional turbulence theorem. The CORE closure concerns the Riemann $C^{(2)}$/MS2 step; turbulence still has to prove its own flux, defect, approximation, gap, and vanishing-viscosity limits.

Paper B is the correct place for the defect theory. Its transfer slots are:

Operator/form. Shell-flux and Littlewood–Paley energy-transfer operators.

Defect. The DFC flux defect, anomalous-dissipation measure, and deviation from the K41 slope.

Approximation. Shell models, DNS, Littlewood–Paley cutoffs, and viscosity families $\nu \to 0$.

Gap. The Hessian gap of the K41 minimiser and the spectral Poincare/SP-TU gap.

Limit. Shell/DNS to PDE, viscosity to zero, and long-time averages to an inertial-range statement.

The most concrete upgrade is to reformulate DFC1 as the dual weighted condition DFC1$^{\vee}$ over sliding scale windows. Then the Kolmogorov 4/5 law, together with the local energy-balance perspective of Duchon–Robert and Eyink, can be used as the exact coarse-grained flux anchor for Paper B, while the remaining projection estimate to Littlewood–Paley weights is kept explicit. The anomalous dissipation results of Brue–De Lellis and collaborators provide the stress-test side: the defect should be represented as a measure or distribution compatible with known vanishing-viscosity scenarios.

5.9 Open directions

Remark 5.4 (Energy cascade and minimax notation)**.**
The Kolmogorov spectrum can be represented by a decoupled minimax penalty: the spectrum minimises the KL free energy while a quadratic penalty enforces constant flux. This is useful notation, not a proved dynamical zero-sum game between injection and dissipation.

Remark 5.5 (Dissipative adaptation)**.**
England's dissipative adaptation principle [27] suggests that driven systems self-organise to maximise the dissipation of absorbed work. In the turbulence context, this provides a thermodynamic rationale for why the cascade spontaneously selects the K41 spectrum: it is the profile that most efficiently channels the injected kinetic energy through the inertial range to the dissipation scales, subject to the flux constraint.

Remark 5.6 (Cumulant expansion for intermittency)**.**
The positivity $\Phi \geq 0$ bounds the cumulant hierarchy of energy fluctuations: $|\kappa_n(E_j)| \leq n!\,\sigma^2/(2\alpha^{n-2})$. This provides a direct route to intermittency exponents ζ_p via the Hessian fluctuations of $F[E]$ around the K41 minimiser.

Remark 5.7 (Shell cascade as Dobrushin influence matrix)**.**
The shell-by-shell energy transfer in the inertial range defines a natural Dobrushin-type influence matrix $C = (c_{jk})$, where c_{jk} quantifies the influence of shell k on shell j via the nonlinear transfer T_j. For the K41 cascade, nearest-neighbour dominance gives $c_{j,j\pm 1} = O(1)$ and $c_{jk} = O(2^{-|j-k|})$ for $|j-k| > 1$. The spectral radius $\rho(C) < 1$, that is, Dobrushin uniqueness, is equivalent to the stability of the K41 minimiser: small perturbations of E^* at shell k decay exponentially in $|j-k|$.

This perspective, imported from the Yang–Mills companion paper where the Dobrushin influence matrix controls the lattice mass gap, reveals a structural isomorphism: the inertial-range cascade is a "one-dimensional lattice system" with exponentially decaying interactions, and K41 is its unique Gibbs state.

Remark 5.8 (DFC as defective monotonicity with controlled defect terms)**.**
The Downhill Flux Condition can be understood as a *defective monotonicity* statement, parallel to the defective Poincaré/LSI approach in the Yang–Mills companion paper [22]:

- *Ideal case (DFC1$^\vee$+DFC2+DFC3):* The free-energy functional $F[E]$ is strictly monotone decreasing along the inertial cascade. K41 is the unique minimiser.
- *Defective case:* Intermittency (bottleneck effect, local backscatter) introduces defect terms δ_j at each shell:

  $$F[E_{j+1}] \;\leq\; F[E_j] - \Delta_j + \delta_j, \qquad \sum_j \delta_j < \infty.$$

  The summability of defects ensures that the cascade remains "asymptotically downhill" despite local violations. This is structurally identical to the defective Dobrushin criterion in Yang–Mills, where the exponentially suppressed defect $e^{-c\sqrt{\beta}}$ (Proposition 3.4 of [22]) plays the same role as the summable δ_j here.

The defective-monotonicity formulation makes the DFC *numerically testable*: measure Δ_j and δ_j at each shell in DNS data, and verify that $\sum \delta_j / \sum \Delta_j \ll 1$. The DFC-with-slack variant is best read as a finite-Reynolds diagnostic until paired with a data-fit or perturbation theorem.

Remark 5.9 (Penalty viewpoint on inertial reductions)**.**
If an inertial-manifold or finite-mode reduction [18] is available, the KL free-energy penalty can be tested in that reduced system with explicit closure errors. Such a reduction may provide a controlled setting for a future coupled game formulation, but the present paper only proves the decoupled penalty statement of Proposition 4.4.

Remark 5.10 (Dobrushin diagnostics on reduced cascades)**.**
The shell cascade as a Dobrushin influence matrix (Remark 5.7) can be tested in finite-mode or inertial reductions when those reductions are available. A verified bound $\rho(C) < 1$ would give a quantitative stability diagnostic for the reduced cascade, but this paper does not prove Dobrushin uniqueness or a turbulence mass gap. The analogy with the Yang–Mills companion paper is therefore a research program, not an imported theorem.

Remark 5.11 (Pattern A Master Stability – cross-problem structure)**.**
This paper instantiates the Pattern A Master Stability template from the unified framework [17]: strict convexity of the renormalised free energy F supplies a local stability margin for the K41 minimiser. A genuine spectral gap for a transfer operator would require the additional cascade-operator analysis left open above.

AI Disclosure

In the preparation of this manuscript, AI-assisted tools (Claude, Anthropic) were used in a supporting capacity, particularly for literature research, text structuring, and formulation assistance. The conceptual design, scientific argumentation, and responsibility for the entire content rest solely with the author.

References

[1] A. N. Kolmogorov, *The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers*, Doklady Akad. Nauk SSSR **30** (1941), 299–303. Reprinted in Proc. R. Soc. Lond. A **434** (1991), 9–13. doi:10.1098/rspa.1991.0075.

[2] A. N. Kolmogorov, *Dissipation of energy in the locally isotropic turbulence*, Doklady Akad. Nauk SSSR **32** (1941), 16–18. Reprinted in Proc. R. Soc. Lond. A **434** (1991), 15–17. doi:10.1098/rspa.1991.0076.

[3] A. N. Kolmogorov, *A refinement of previous hypotheses concerning the local structure of turbulence in a viscous incompressible fluid at high Reynolds number*, J. Fluid Mech. **13** (1962), 82–85. doi:10.1017/S0022112062000518.

[4] L. Onsager, *Statistical hydrodynamics*, Nuovo Cimento Suppl. **6** (1949), 279–287. doi:10.1007/BF02780991.

[5] P. Constantin, W. E, and E. S. Titi, *Onsager's conjecture on the energy conservation for solutions of Euler's equation*, Comm. Math. Phys. **165** (1994), 207–209. doi:10.1007/BF02099744.

[6] J. Duchon and R. Robert, *Inertial energy dissipation for weak solutions of incompressible Euler and Navier–Stokes equations*, Nonlinearity **13** (2000), 249–255. doi:10.1088/0951-7715/13/1/312.

[7] P. Isett, *A proof of Onsager's conjecture*, Ann. of Math. **188** (2018), 871–963. doi:10.4007/annals.2018.188.3.4.

[8] Z.-S. She and E. Lévêque, *Universal scaling laws in fully developed turbulence*, Phys. Rev. Lett. **72** (1994), 336–339. doi:10.1103/PhysRevLett.72.336.

[9] G. L. Eyink and K. R. Sreenivasan, *Onsager and the theory of hydrodynamic turbulence*, Rev. Mod. Phys. **78** (2006), 87–135. doi:10.1103/RevModPhys.78.87.

[10] U. Frisch, *Turbulence: The Legacy of A. N. Kolmogorov*, Cambridge University Press, 1995. doi:10.1017/CBO9781139170666.

[11] G. L. Eyink, *Local* 4/5*-law and energy dissipation anomaly in turbulence*, Nonlinearity **16** (2003), 137–145. doi:10.1088/0951-7715/16/1/309.

[12] R. H. Kraichnan, *Inertial ranges in two-dimensional turbulence*, Phys. Fluids **10** (1967), 1417–1423. doi:10.1063/1.1762301.

[13] A. S. Monin and A. M. Yaglom, *Statistical Fluid Mechanics*, Vols. 1–2, MIT Press, 1971–1975.

[14] Y. Kaneda, T. Ishihara, M. Yokokawa, K. Itakura, and A. Uno, *Energy dissipation rate and energy spectrum in high resolution direct numerical simulations of turbulence in a periodic box*, Phys. Fluids **15** (2003), L21–L24. doi:10.1063/1.1539855.

[15] T. Gotoh, D. Fukayama, and T. Nakano, *Velocity field statistics in homogeneous steady turbulence obtained using a high-resolution direct numerical simulation*, Phys. Fluids **14** (2002), 1065–1081. doi:10.1063/1.1448296.

[16] S. B. Pope, *Turbulent Flows*, Cambridge University Press, 2000. doi:10.1017/CBO9780511840531.

[17] L. Geiger, *The Renormalized Free Energy Framework: A Unified Resolution of Structural Bottlenecks*, Zenodo, March 2026. doi:10.5281/zenodo.19036190.

[18] C. Foias, G. R. Sell, and R. Temam, *Inertial manifolds for nonlinear evolutionary equations*, J. Differential Equations **73** (1988), 309–353. doi:10.1016/0022-0396(88)90110-6.

[19] T. D. Drivas, *Turbulent cascade direction and Lagrangian time-asymmetry*, J. Nonlinear Sci. **29** (2019), 65–88. doi:10.1007/s00332-018-9476-8; arXiv:1802.02289.

[20] L. De Rosa, T. D. Drivas, and M. Inversi, *On the support of anomalous dissipation measures*, J. Math. Fluid Mech. **26** (2024), Art. 56. doi:10.1007/s00021-024-00894-z; arXiv:2301.09603.

[21] D. Albritton, E. Brué, and M. Colombo, *Non-uniqueness of Leray solutions of the forced Navier–Stokes equations*, Ann. of Math. **196** (2022), 415–455. doi:10.4007/annals.2022.196.1.3.

[22] L. Geiger, *Volume-Independent Spectral Gap for Strong-Coupling Lattice Gauge Theory via Poincaré-Dobrushin Machinery*, Preprint, 2026. doi:10.5281/zenodo.19087433.

[23] S. G. Saddoughi and S. V. Veeravalli, *Local isotropy in turbulent boundary layers at high Reynolds number*, J. Fluid Mech. **268** (1994), 333–372. doi:10.1017/S0022112094001370.

[24] Y. Li, E. Perlman, M. Wan, Y. Yang, C. Meneveau, R. Burns, S. Chen, A. Szalay, and G. Eyink, *A public turbulence database cluster and applications to study Lagrangian evolution of velocity increments in turbulence*, J. Turbulence **9** (2008), N31. doi:10.1080/14685240802376389.

[25] V. S. L'vov, E. Podivilov, A. Pomyalov, I. Procaccia, and D. Vandembroucq, *Improved shell model of turbulence*, Phys. Rev. E **58** (1998), 1811–1822. doi:10.1103/PhysRevE.58.1811.

[26] T. Buckmaster and V. Vicol, *Nonuniqueness of weak solutions to the Navier–Stokes equation*, Ann. of Math. **189** (2019), 101–144. doi:10.4007/annals.2019.189.1.3.

[27] J. L. England, *Dissipative adaptation in driven self-assembly*, Nature Nanotechnology **10** (2015), 919–923. doi:10.1038/nnano.2015.250.